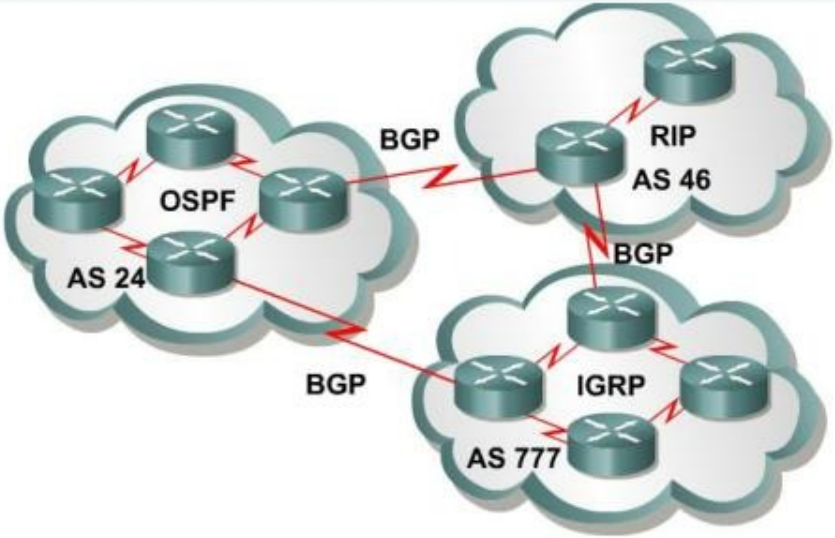
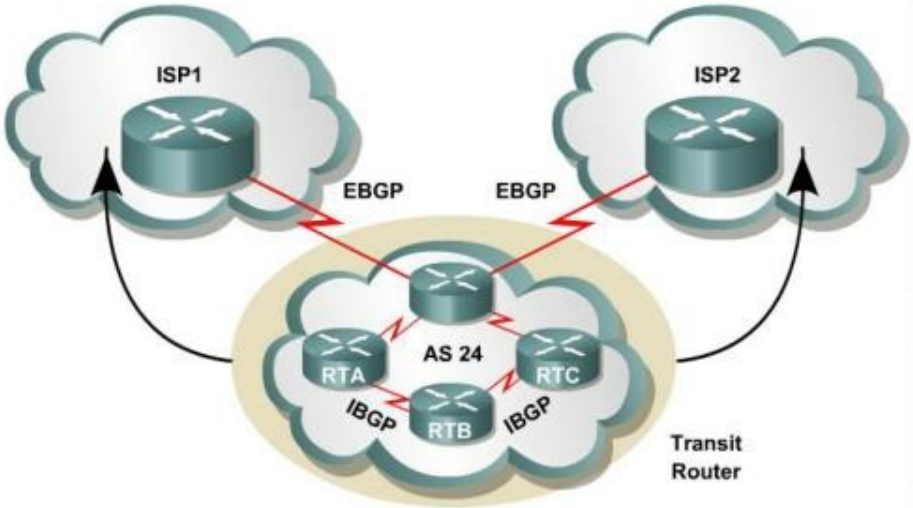
BGP Training

Terms

- **IGP** (Interior Gateway Protocol) - RIP, IGRP, EIGRP, OSPF = Routing protocol used to exchange routing information within an autonomous system.
- **EGP** (Exterior Gateway Protocol) - BGP = Routing protocol used to exchange routing information between autonomous systems.
- **Autonomous System** = (From RFC 1771) “A set of routers under the single technical administration, using an IGP and common metrics to route packets within the AS, and using an EGP to route packets to other AS’s.”
- **BGP** is a path vector or an advanced distance vector routing protocol.

When to use BGP and when not to use BGP

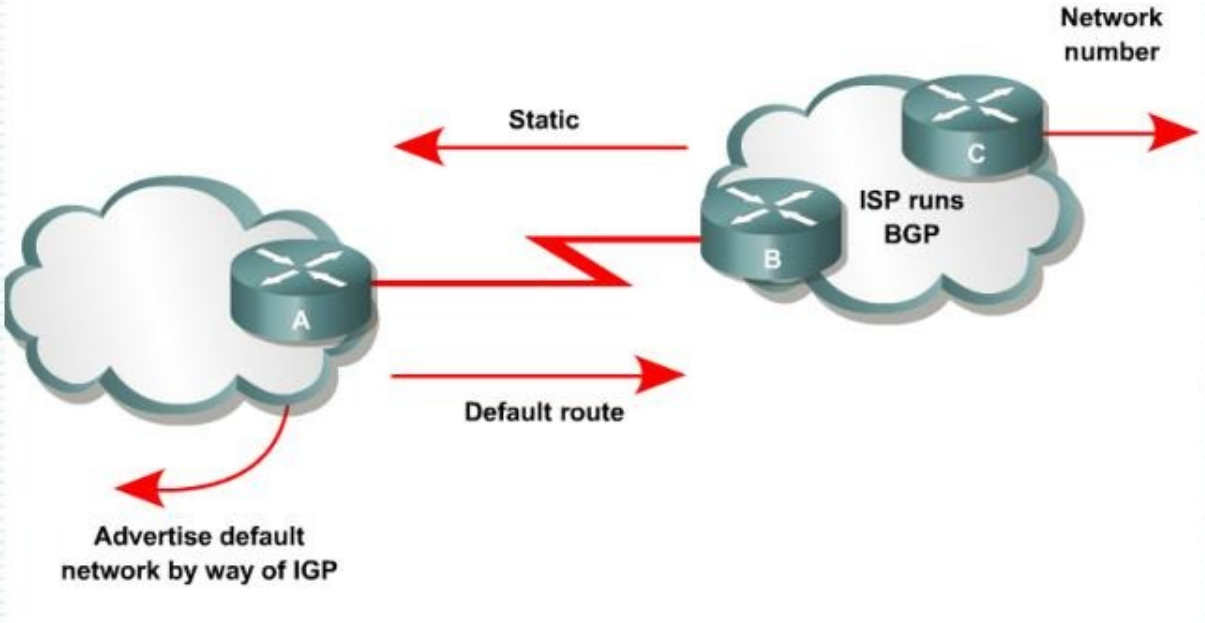




Use BGP when the effects of BGP are well understood and one of the following conditions exist:

- The AS allows packets to transit through it to reach another AS (transit AS).
- The AS has multiple connections to other AS's.
- The flow of traffic entering or exiting the AS must be manipulated. This is policy based routing and based on **attributes**.

When to use BGP and when not to use BGP

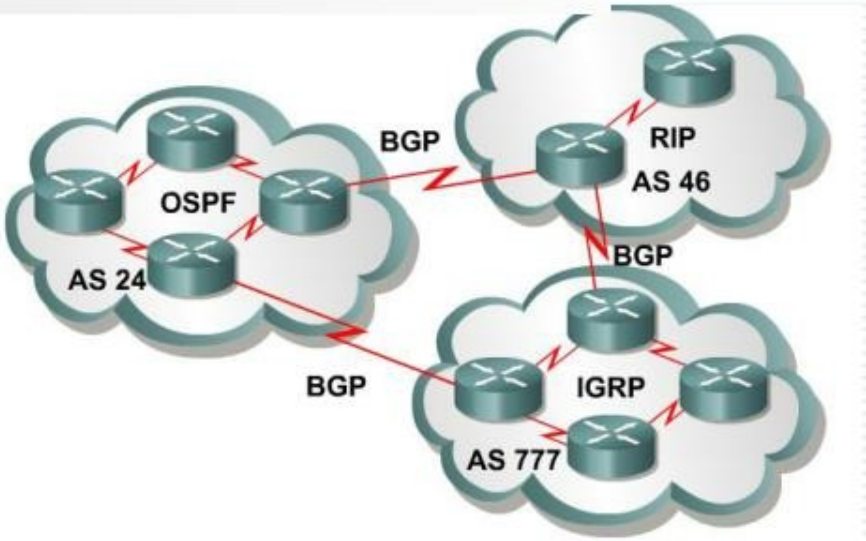


Do not use BGP if you have one or more of the following conditions:

- A single connection to the Internet or another AS
- No concern for routing policy or routing selection
- A lack of memory or processing power on your routers to handle constant BGP updates
- A limited understanding of route filtering and BGP path selection process
- Low bandwidth between AS's

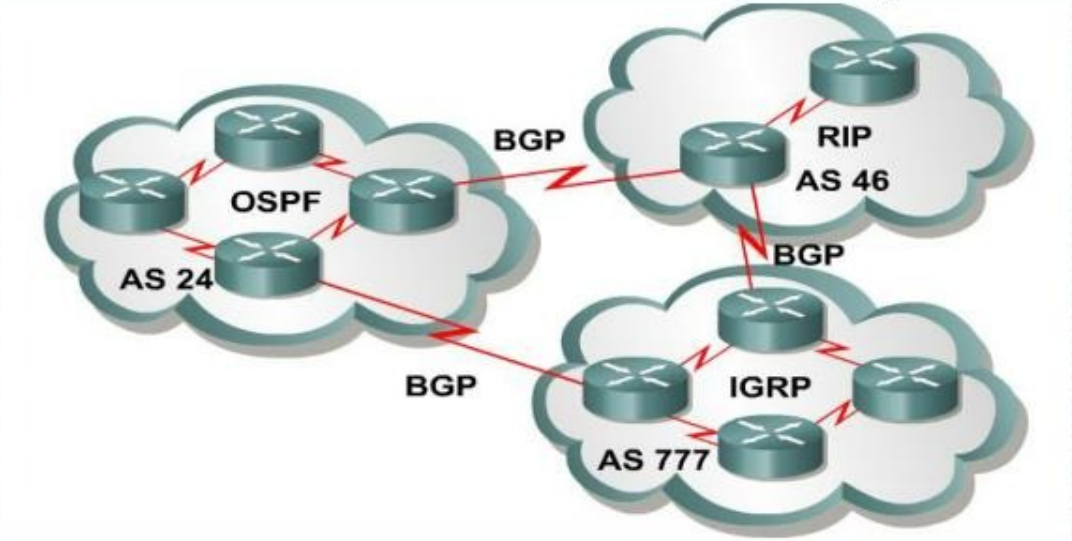
Overview of autonomous systems

EGPs, such as BGP, are used to interconnect autonomous systems.



- An AS is a group of routers that share similar routing policies and operate within a single administrative domain.
- An AS can be a collection of routers running a single IGP, or it can be a collection of routers running different protocols all belonging to one organization.
- In either case, the outside world views the entire Autonomous System as a single entity.

Overview of autonomous systems

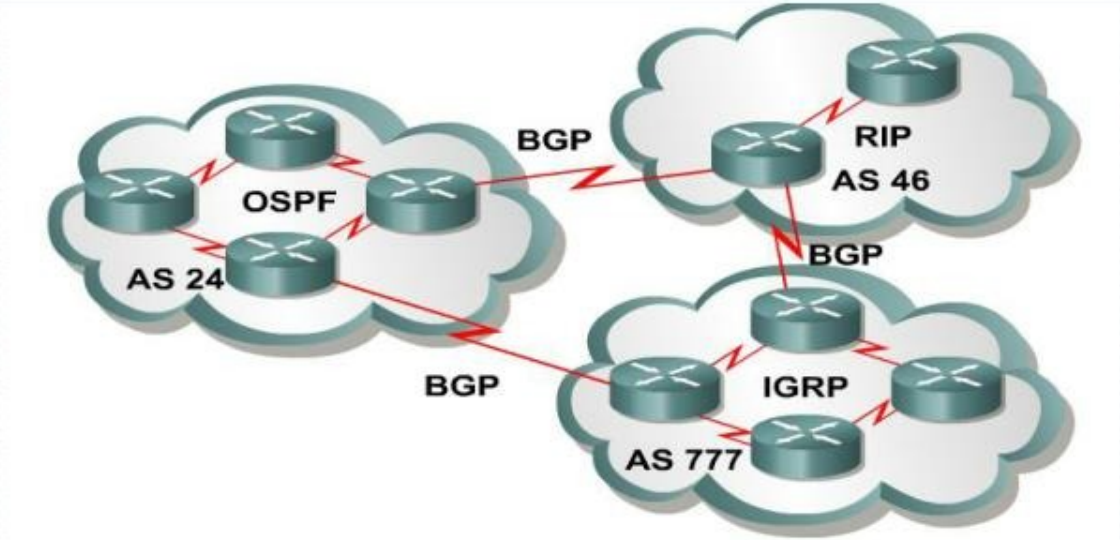


AS Numbers

- Each AS has an identifying number that is assigned by an Internet registry or a service provider.
- This number is between **1 and 65,535**.
- AS numbers within the range of **64,512 through 65,535** are reserved for **private** use.
- This is similar to RFC 1918 IP addresses.
- Because of the finite number of available AS numbers, an organization must present justification of its need before it will be assigned an AS number.

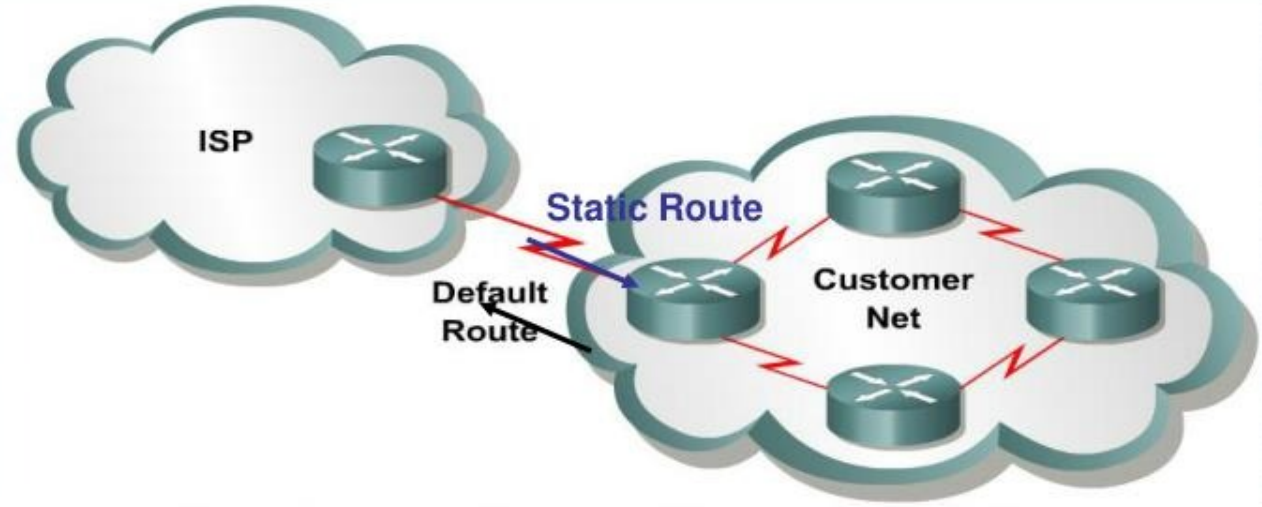
HCL

Overview of autonomous systems



- Today, the Internet Assigned Numbers Authority (IANA) is enforcing a policy whereby organizations that connect to a single provider and share the provider's routing policies use an AS number from the private pool, 64,512 to 65,535.

Single-homed autonomous systems

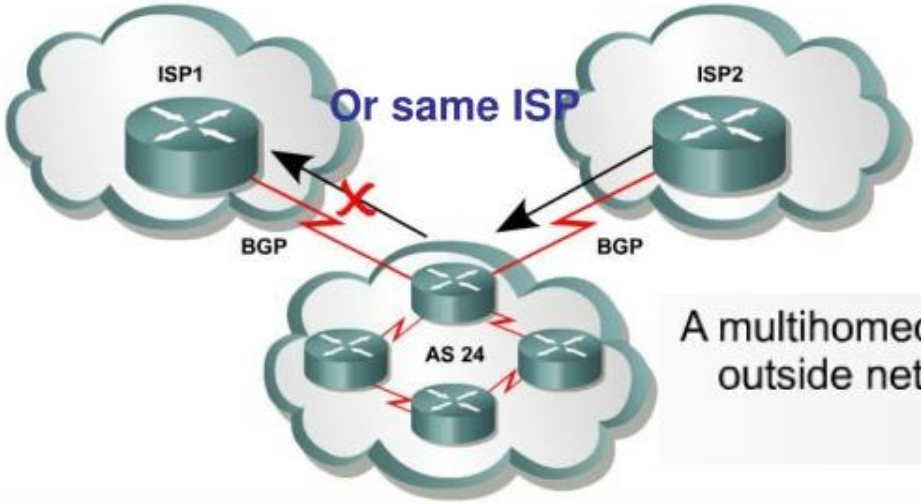


A single-homed AS can be configured with a default route to reach outside networks.

- If an AS has only **one exit point** to outside networks, it is considered a **single-homed system**.
- Single-homed autonomous systems are often referred to as **stub** networks or stubs.
- Stubs can rely on a **default route** to handle all traffic destined for non-local networks.
- BGP is **not** normally needed in this situation.

HCL

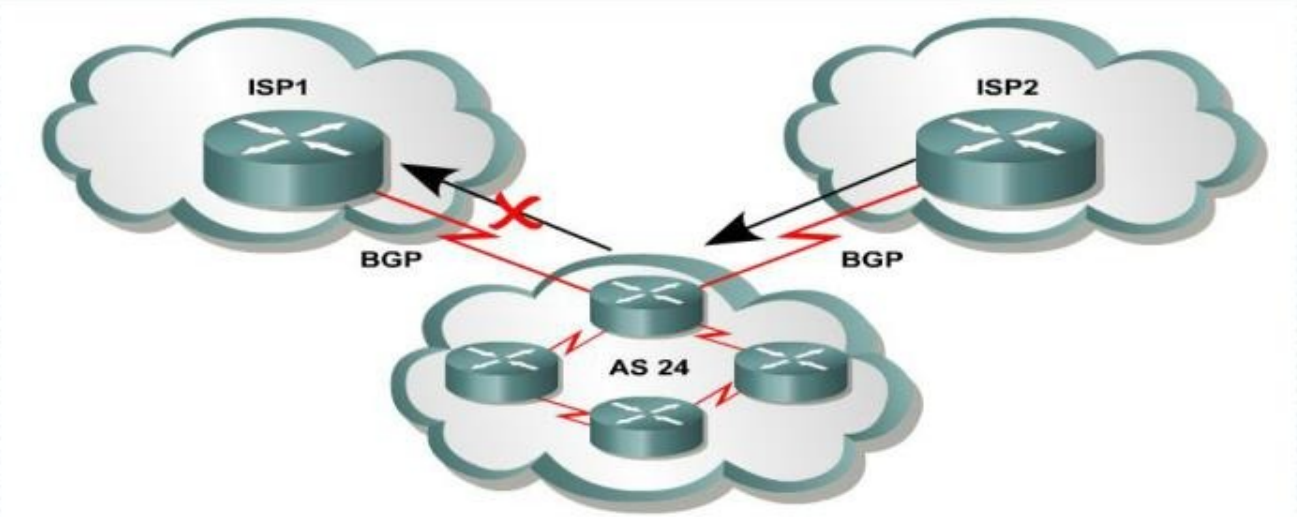
Multihomed nontransit autonomous systems



A multihomed nontransit AS features more than one exit point to outside networks, but does not allow traffic to pass from one outside connection to another.

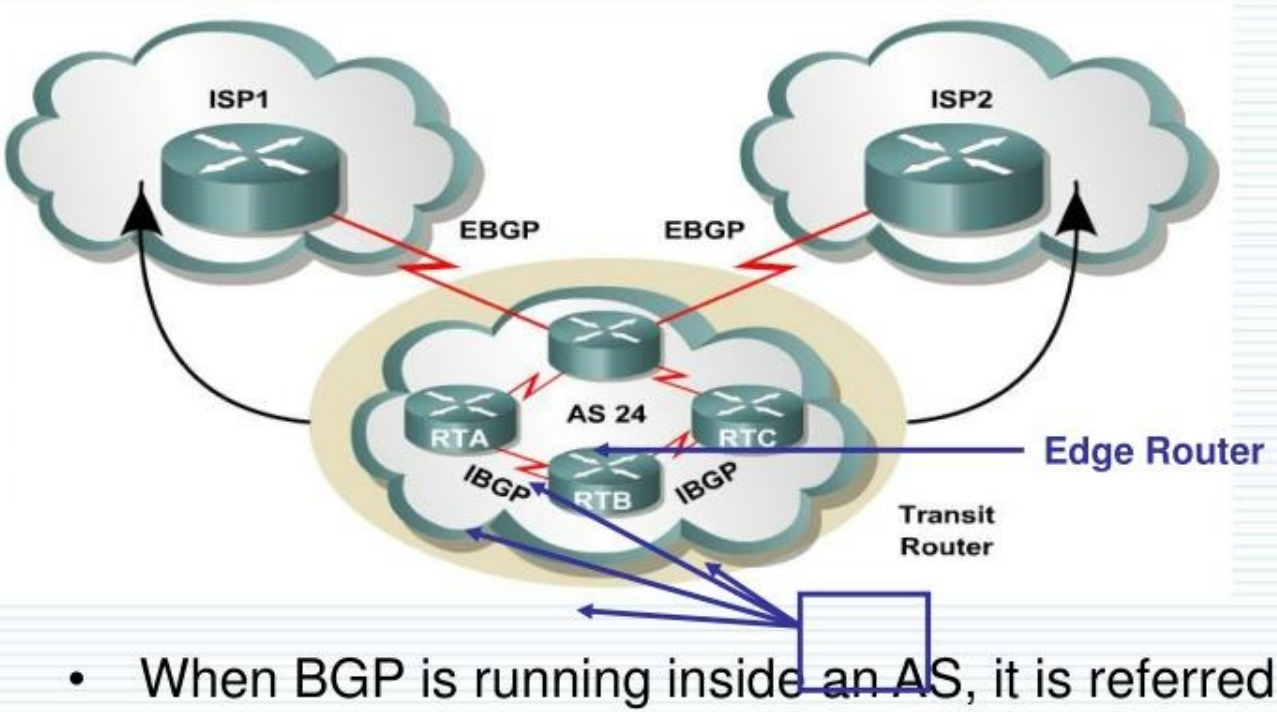
- Multihomed nontransit autonomous systems do not really need to run BGP4 with their providers.
- It is usually **recommended** and **often required** by ISPs.
- As it will be seen later in this module, BGP4 offers numerous advantages, including increased control of route propagation and filtering.

Multihomed nontransit autonomous systems



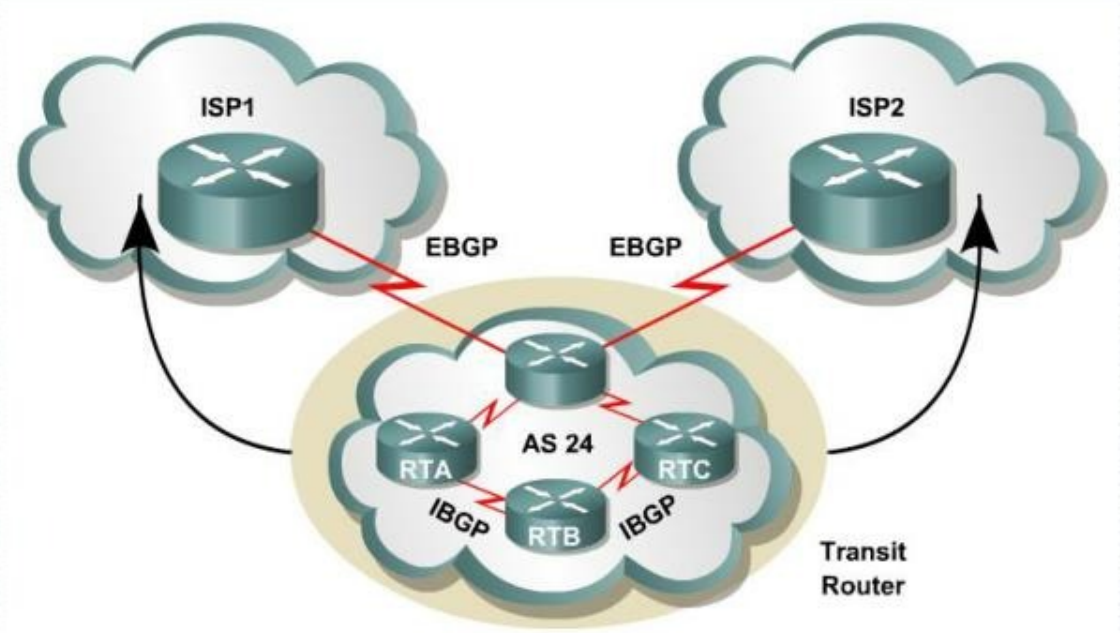
- ***Incoming route advertisements influence your outgoing traffic, and outgoing advertisements influence your incoming traffic.***
- If the provider advertises routes into your AS via BGP, your internal routers have more accurate information about external destinations.
  - BGP also provides tools for setting routing policies for external destinations.
- If your internal routes are advertised to the provider via BGP, you have influence over which routes are advertised at which exit point.
  - BGP also provides tools for your influencing (to some degree) the choices the provider makes when sending traffic into your AS.

Multi-homed Transit Autonomous Systems



- When BGP is running inside an AS, it is referred to as **Internal BGP (IBGP).**
- When BGP runs between autonomous systems, it is called **External BGP (EBGP).**
- If the role of a BGP router is to route IBGP traffic, it is called a **transit router.**
- Routers that sit on the boundary of an AS and that use EBGP to exchange information with the ISP are called **border or edge routers**.

BGP Basics



- BGP is a path vector routing protocol.
- Defined in RFC 1772
- BGP is a distance vector routing protocol, in that it relies on downstream neighbors to pass along routes from their routing table.
- BGP uses a list of AS numbers through which a packet must pass to reach a destination.

BGP Message Types

- Before establishing a BGP peer connection the two neighbors must perform the standard TCP three-way handshake and open a TCP connection to port 179.
- After the TCP session is established, BGP peers exchanges several messages to open and confirm connection parameters and to send BGP routing information.
- All BGP messages are unicast to the one neighbor over the TCP connection.
- There are four BGP message types:
  - **Type 1: OPEN**
  - **Type 2: KEEPALIVE**
  - **Type 3: UPDATE**
  - **Type 4: NOTIFICATION**

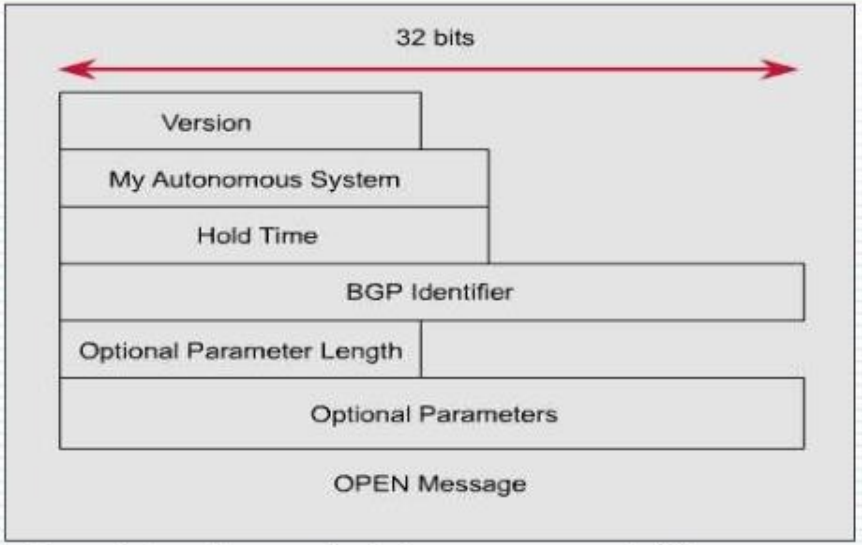
BGP Message Types

Each BGP Message contains the following header:

- **Marker**: The marker field is used to either authenticate incoming BGP messages or to detect loss of synchronization between two BGP peers.
- **Length**: The length field indicates the total BGP message length, including the header.



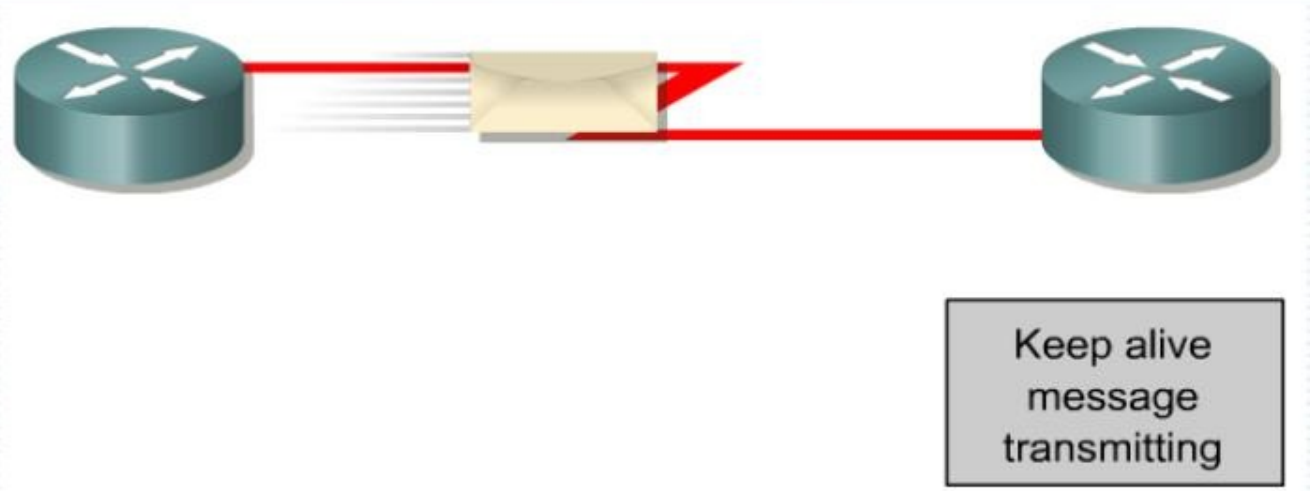
Type 1:
BGP Open Message



- After the TCP session is established, both neighbors send Open messages.
- This message is used to establish connections with peers.
- Each neighbor uses this message to identify itself and to specify its BGP operational parameters including:
  - **BGP version number** (defaults to version 4)
  - **AS number**: AS number of the originating router, determines if BGP session is EBGP or IBGP.
  - **BGP identifier**: IP address that identifies the neighbor using the same method as OSPF router ID.
  - **Optional parameter**: authentication, multiprotocol support and route refresh.

HCL

Type 2: BGP Keepalive Message



- This message type is sent periodically between peers to maintain connections and verify paths held by the router sending the keepalive.
- If a router accepts the parameters specified in its neighbor's Open message, it responds with a Keepalive.
- Subsequent Keepalives are **sent every 60 seconds** by Cisco default or equal to one-third the agreed-upon hold time (180 seconds).
- If the periodic timer is set to a value of zero (0), no keepalives are sent.

Type 3: BGP Update Message



- The UPDATE messages contain all the information BGP uses to construct a loop-free picture of the internetwork.
- Update messages advertises feasible routes, withdrawn routes, or both.
- The three basic components of an UPDATE message are:
  - **Network-Layer Reachability Information (NLRI)**
  - **Path Attributes**
  - **Withdrawn Routes**

Type 3: BGP Update Message

Network-Layer Reachability Information (NLRI)

- This is one or more (Length, Prefix) tuples that advertise IP address prefixes and their lengths.
- 192.168.160.0/19
  - Prefix = 192.168.160.0
  - Prefix Length = 19

Path Attributes

- This is described later, providing the information that allows BGP to choose a shortest path, detect routing loops, and determine routing policy.

Withdrawn Routes

- These are (Length, Prefix) tuples describing destination that have become unreachable and are being withdrawn from service.
- An update message that has no NLRI or path attribute information is used to advertise only routes to be withdrawn from service.

Path Attributes

Attribute Code	Type
1 — ORIGIN	Well-known mandatory
3 — NEXT_HOP	Well-known mandatory
2 — AS_PATH	Well-known mandatory
4 — MULTI_EXIT_DISC	Optional nontransitive
5 — LOCAL_PREF	Well-known discretionary
6 — ATOMIC_AGGREGATE	Well-known discretionary
7 — AGGREGATOR	Well-known discretionary
8 — COMMUNITY	Optional transitive (Cisco)
9 — ORIGINATOR_ID	Optional nontransitive (Cisco)
10 — Cluster List	Optional nontransitive (Cisco)
11 — Destination Preference	(MCI)
12 — Advertiser	(Baynet)
13 — rcid_path	(Baynet)
255 — Reserved	—

Optional transitive

- An attribute that may or may not be, recognized by all BGP implementations (thus, **optional**).
- Because the attribute is **transitive**, BGP **should accept and advertise** the attribute even if it isn't recognized.

Example: COMMUNITY

Path Attributes

Attribute Code	Type
1 — ORIGIN	Well-known mandatory
3 — NEXT_HOP	Well-known mandatory
2 — AS_PATH	Well-known mandatory
4 — MULTI_EXIT_DISC	Optional nontransitive
5 — LOCAL_PREF	Well-known discretionary
6 — ATOMIC_AGGREGATE	Well-known discretionary
7 — AGGREGATOR	Well-known discretionary
8 — COMMUNITY	Optional transitive (Cisco)
9 — ORIGINATOR_ID	Optional nontransitive (Cisco)
10 — Cluster List	Optional nontransitive (Cisco)
11 — Destination Preference	(MCI)
12 — Advertiser	(Baynet)
13 — rcid_path	(Baynet)
255 — Reserved	—

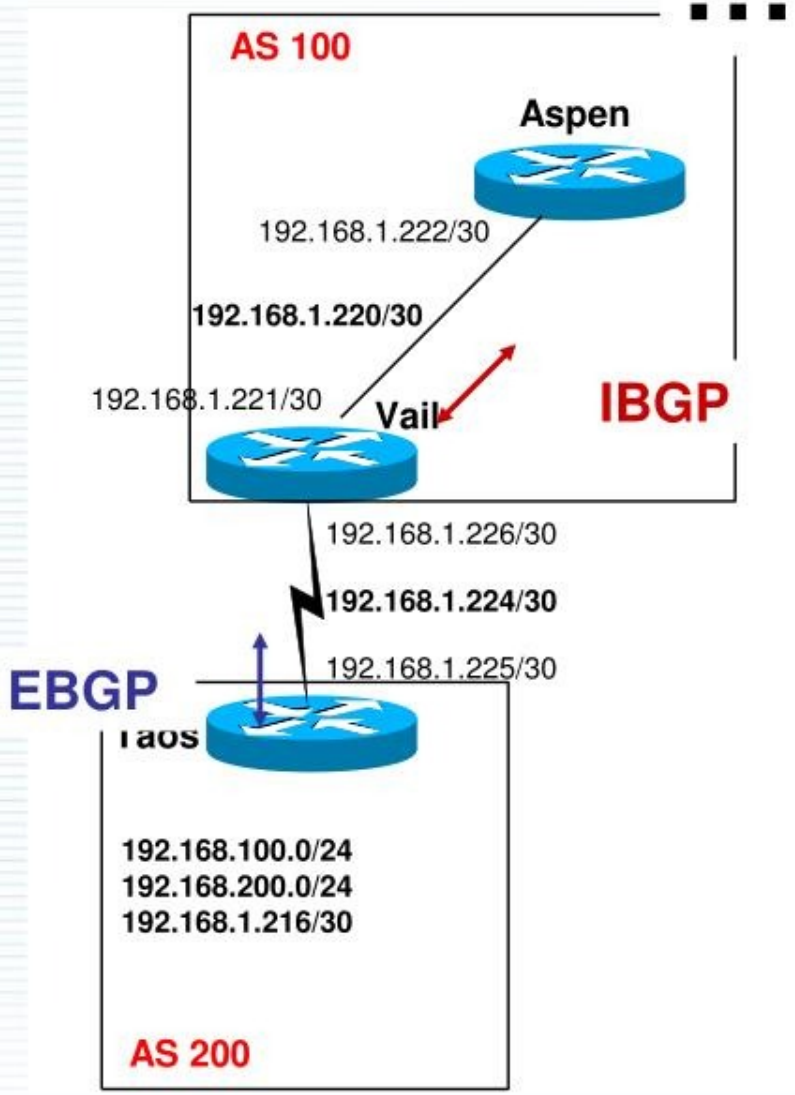
Optional non-transitive

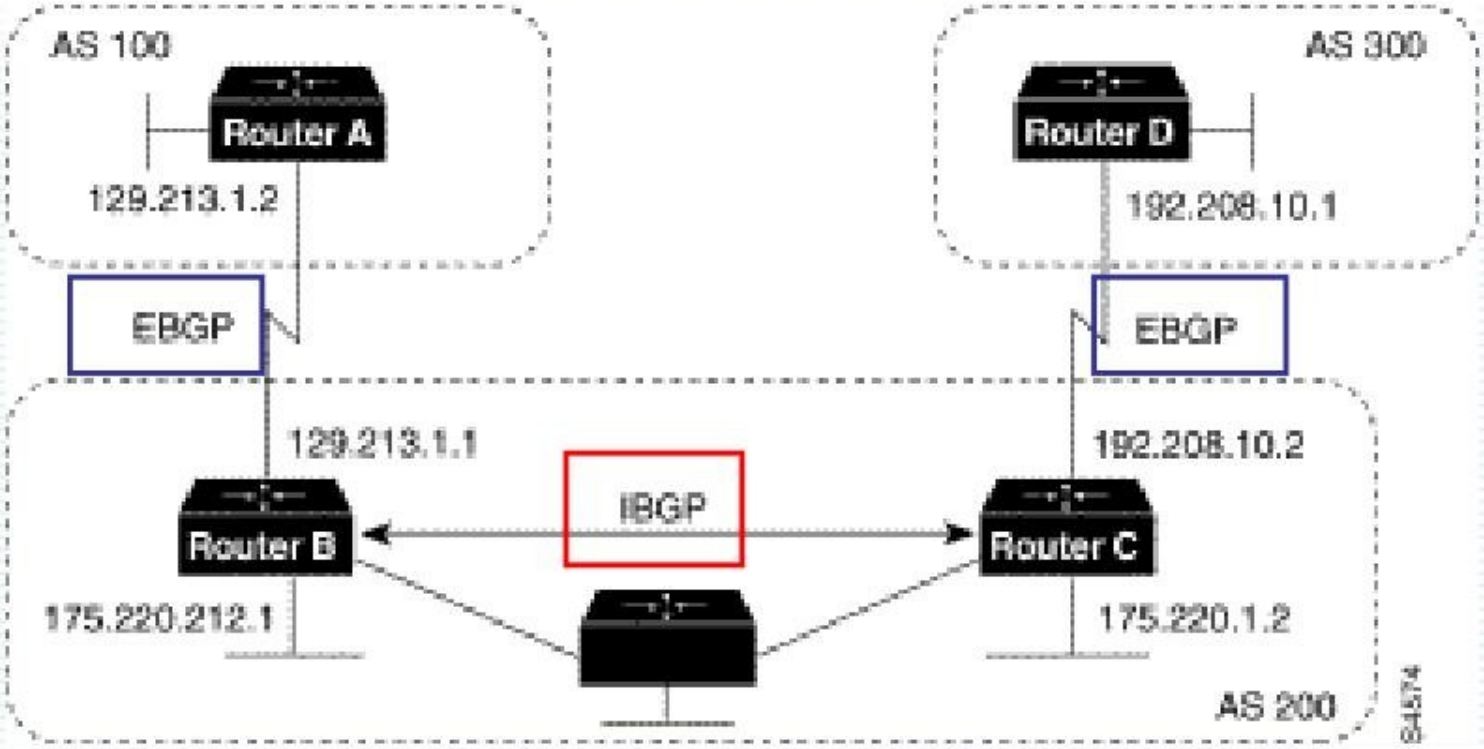
- An attribute that **may or may not be, recognized** by all BGP implementations.
- Whether or not the receiving BGP router recognizes the attribute, it is **non-transitive**, and **should not be passed along to other BGP peers**.

Example: ORIGINATOR_ID

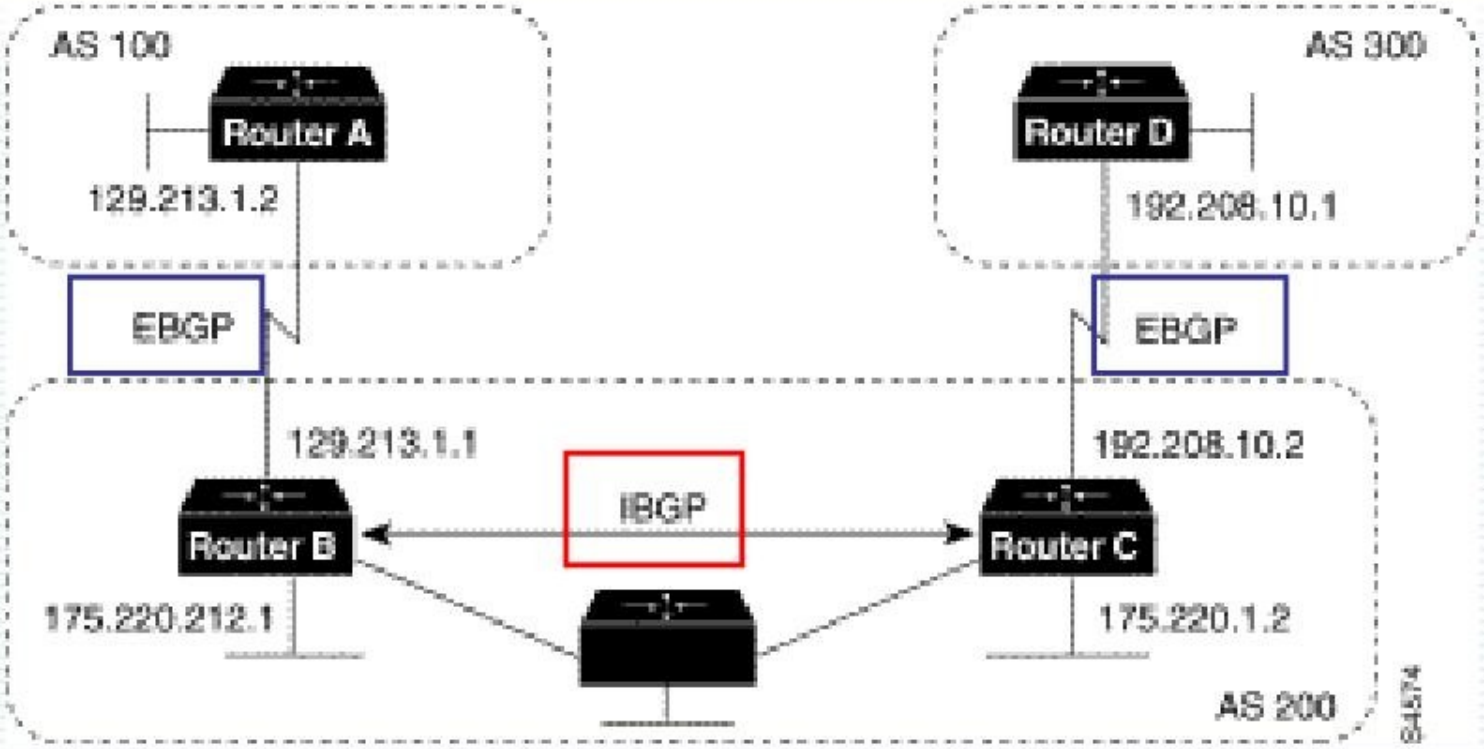
IBGP vs EBGP

- When BGP is running inside an AS, it is referred to as **Internal BGP (IBGP)**.
  - If a BGP router's role is to route IBGP traffic, it is called a transit router.
- When BGP runs between autonomous systems, it is called **External BGP (EBGP)**.
  - Routers that sit on the boundary of an AS and use EBGP to exchange information with the ISP are called border routers.



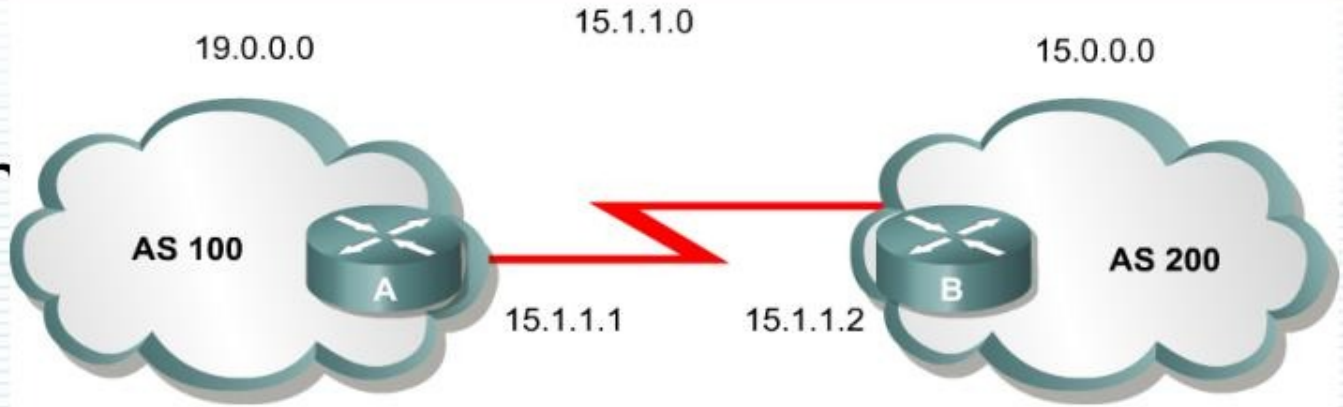


- Routers A and B are running **EBGP (BGP)**, and Routers B and C are running **IBGP**.
- Note that the **EBGP (BGP)** peers are directly connected and that the **IBGP** peers are not. (They can be.)
- As long as there is an **IGP** running that allows the two neighbors to reach one another, IBGP peers do not have to be directly connected.



- All **BGP** speakers within an AS must establish a peer relationship with each other, that is, the **BGP** speakers within an AS must be fully meshed logically. (later)
- BGP4 provides two techniques that alleviate the requirement for a logical full mesh: confederations and route reflectors. (later)
- AS 200 is a ***transit AS*** for AS 100 and AS 300---that is, AS 200 is used to transfer packets between AS 100 and AS 300.

BGP Configuration



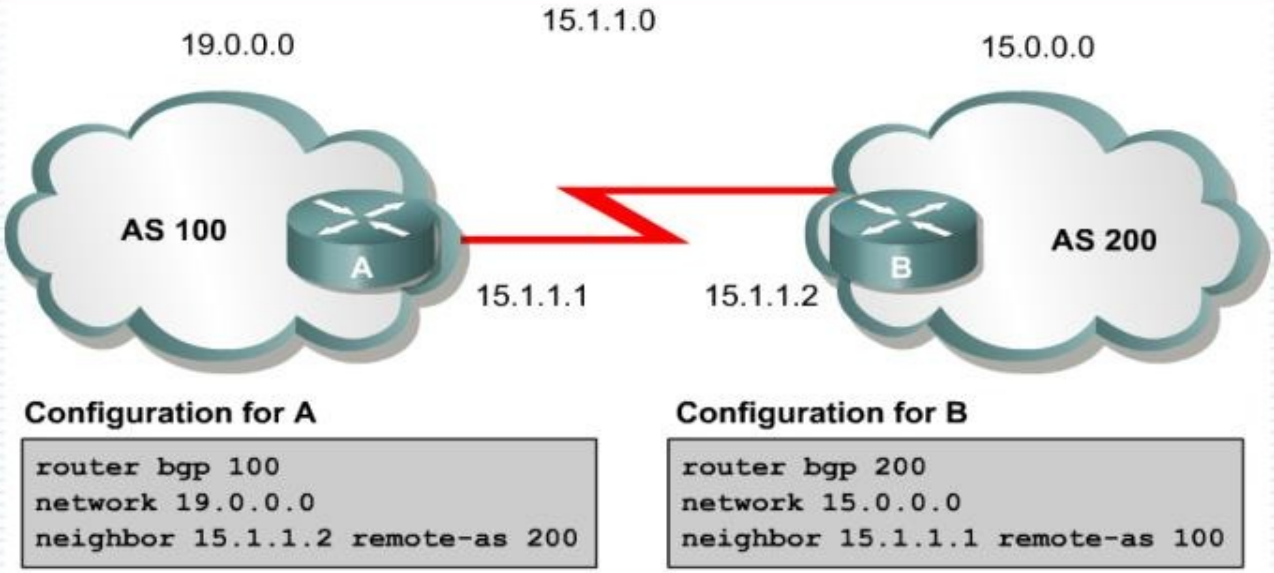
- To begin configuring a BGP process, issue the following familiar command:

```
Router(config)#router bgp AS-number
```

- BGP configuration commands appear on the surface to mirror the syntax of familiar IGP (for example, RIP, OSPF) commands.
- Although the syntax is similar, the function of these commands is significantly different.
- **Note**: Cisco IOS permits only one BGP process to run at a time, thus, **a router cannot belong to more than one AS.**

HCL

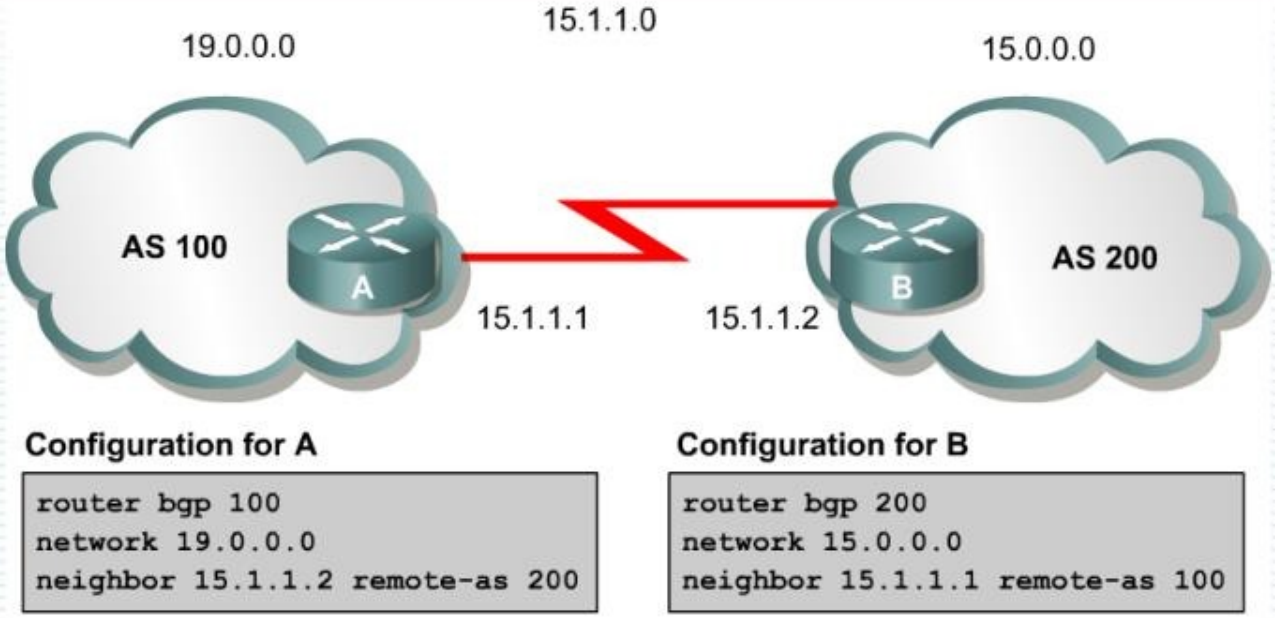
BGP Configuration



`Router(config-router)#`**network *network-number* [mask *network-mask*]**

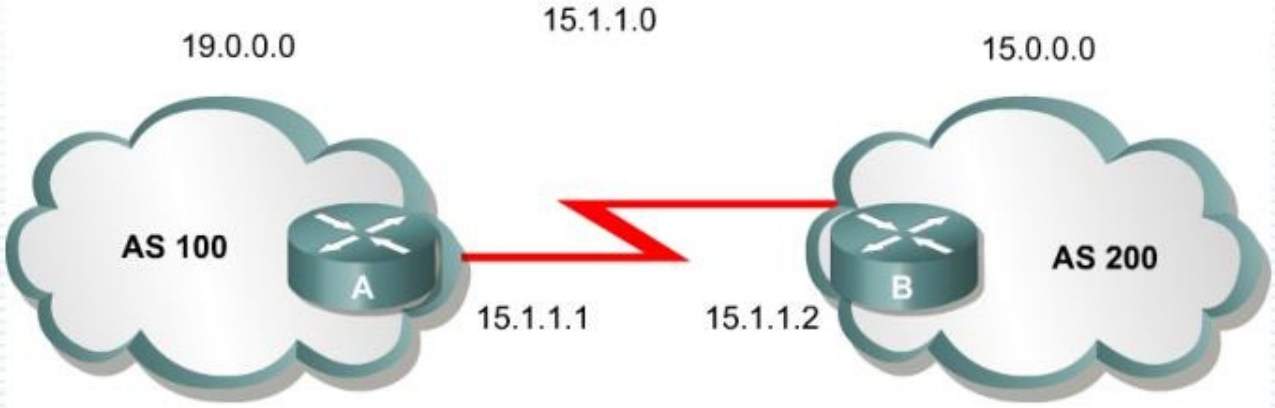
- The **network** command is used with **IGPs**, such as RIP, to determine the interfaces on which to send and receive updates, as well as which directly connected networks to advertise.
- However, when configuring **BGP**, the **network** command does **not** affect what interfaces BGP runs on.
- In BGP, the **network** command tells the BGP process **what locally learned networks to advertise**. .

BGP Configuration



- ***These networks must also exist in the local router's routing table (show ip route), or they will not be sent out in updates.***
- You can use the **mask** keyword with the **network** command to specify individual subnets.
- Routes learned by the BGP process are propagated by default, but are often filtered by a routing policy.

BGP Configuration

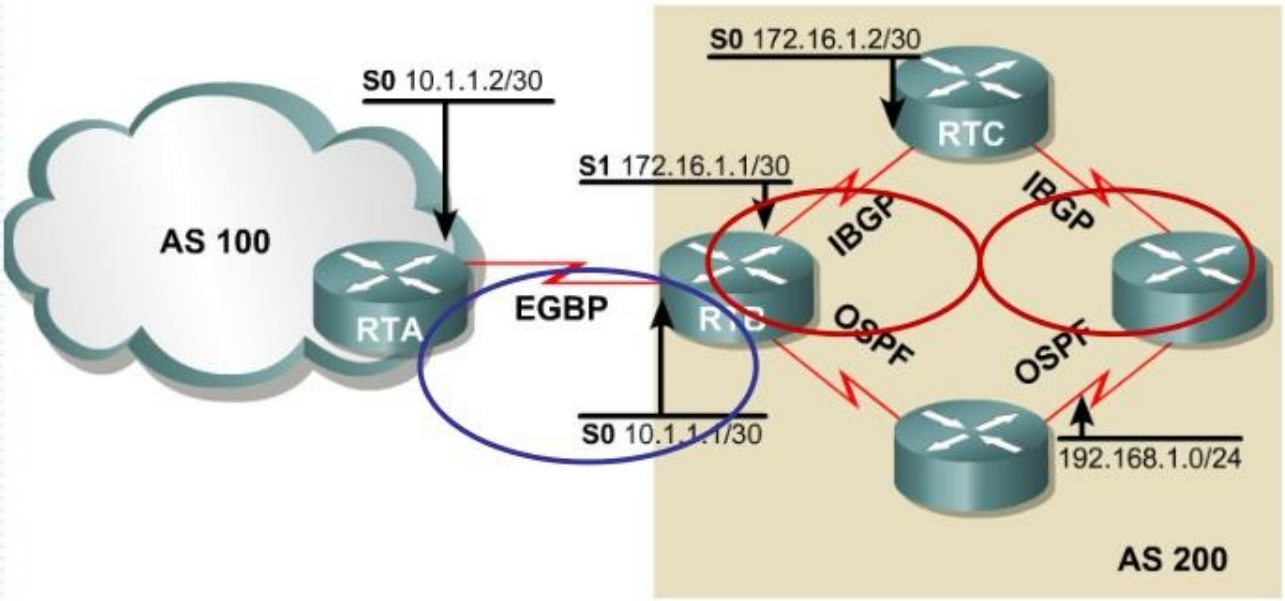


`Router(config-router)#` **`neighbor` *`ip-address`* `remote-as` *`AS-number`***

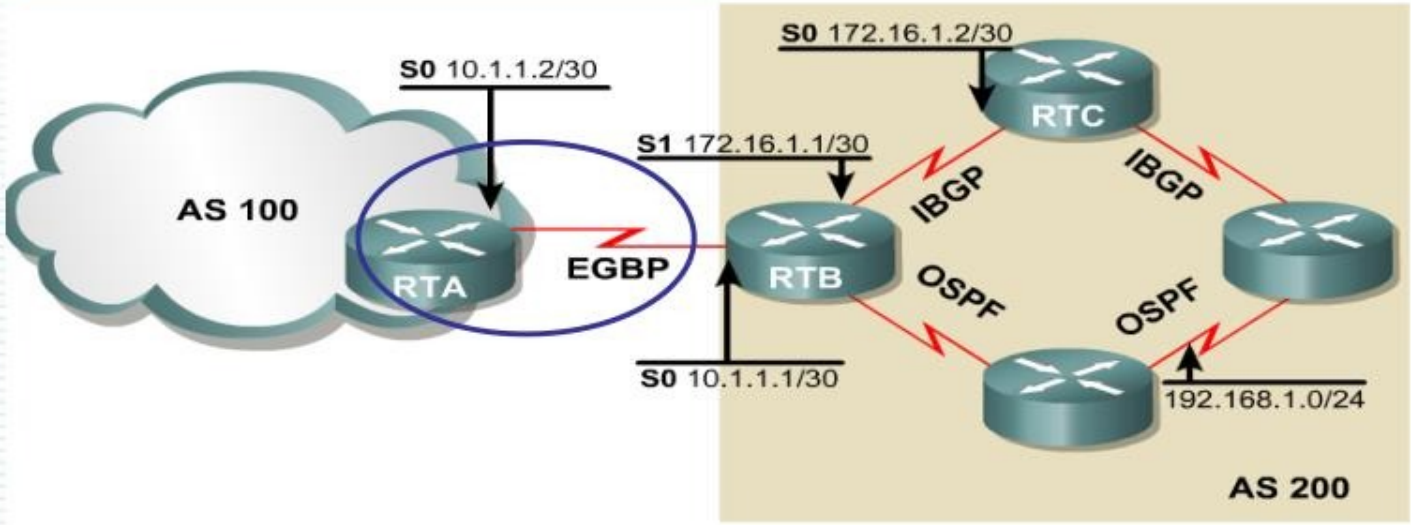
- In order for a BGP router to **establish a neighbor relationship with another BGP router**, you must issue the this configuration command.
- This command serves to identify a peer router with which the local router will establish a session.
- The ***AS-number*** argument determines whether the neighbor router is an EBGP or an IBGP neighbor.

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BGP Configuration



- If the ***AS-number*** configured in the **router bgp** command is **identical** to the ***AS-number*** configured in the **neighbor** statement, BGP will initiate an **internal session - IBGP**.
- If the field values are **different**, BGP will build an **external session - EBGP**.

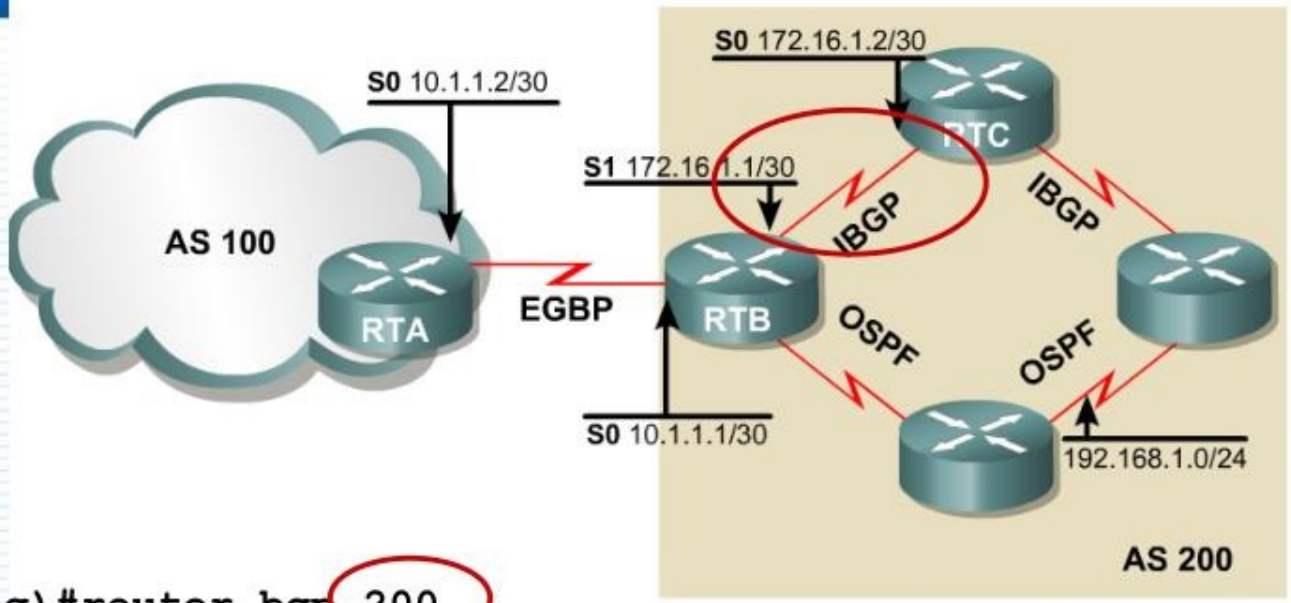


EBGP

```
RTA(config)#router bgp 100
RTA(config-router)#neighbor 10.1.1.1 remote-as 200

RTB(config)#router bgp 200
RTB(config-router)#neighbor 10.1.1.2 remote-as 100
```

- RTB: Note that the **neighbor** command's **remote-as** value, 100, is different from the AS number specified by the **router bgp** command (200).
- Because the two AS numbers are different, BGP will start an **EBGP** connection with RTA.
- Communication will occur between autonomous systems.

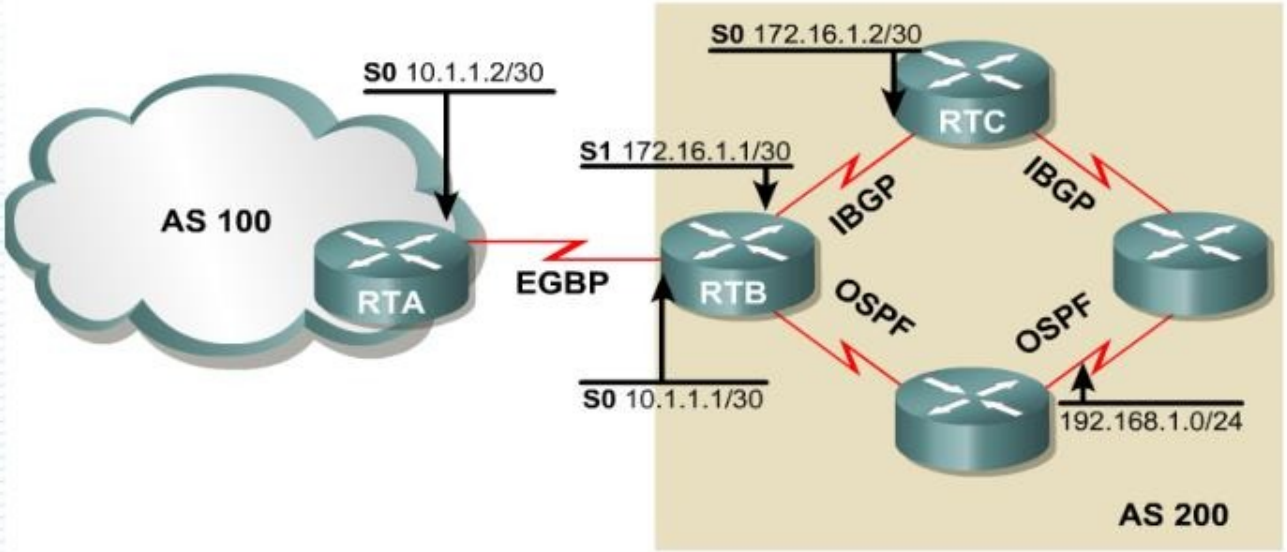


IBGP

```
RTB(config)#router bgp 200
RTB(config-router)#neighbor 172.16.1.2 remote-as 200
RTB(config-router)#neighbor 172.16.1.2 update-source loopback 0

RTC(config)#router bgp 200
RTC(config-router)#neighbor 172.16.1.1 remote-as 200
RTC(config-router)#neighbor 172.16.1.1 update-source loopback 0
```

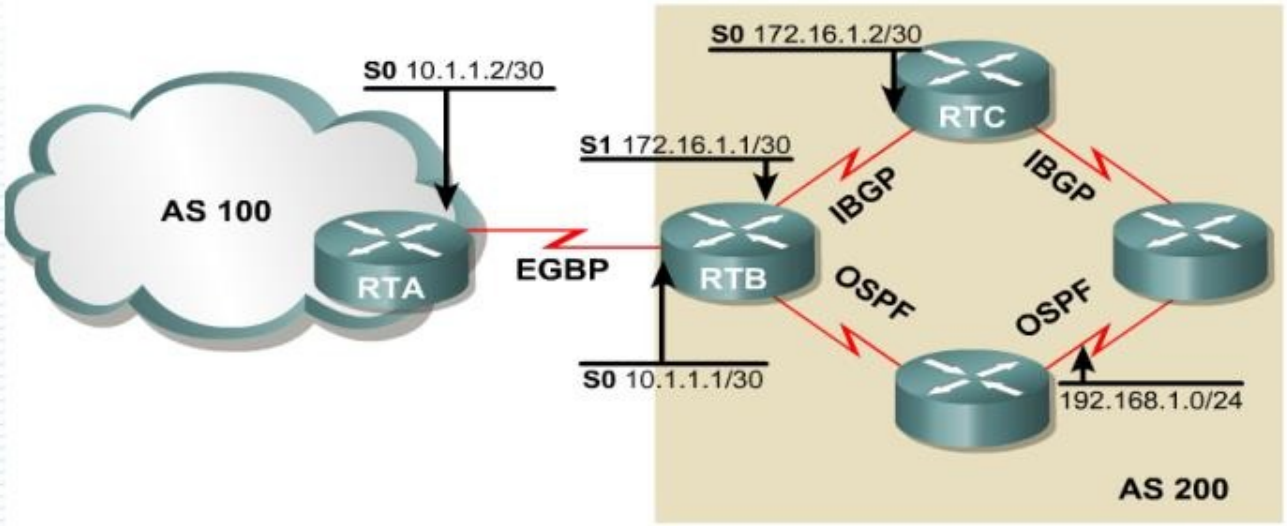
- Since the **remote-as** value (200) is the same as RTB's BGP AS number, BGP recognizes that this connection will occur within AS 200, so it attempts to establish an IBGP session.
- In reality, AS 200 is not a remote AS at all; it is the local AS, since both routers live there. But for simplicity, the keyword **remote-as** is used when configuring both EBGP and IBGP sessions.




```
RTB(config-router)#neighbor 172.16.1.2 update-source loopback 0
RTC(config-router)#neighbor 172.16.1.1 update-source loopback 0
```

- The **update-source loopback 0** command is used to instruct the router to use *any* operational interface for TCP connections (as long as Lo0 is up and configured with an IP address).
- Without the **update-source loopback 0** command, BGP routers can use only the closest IP interface to the peer.
- The ability to use any operational interface provides BGP with robustness in the event the link to the closet interface fails.
  - Since EBGP sessions are typically point-to-point, there is no need to use this command with EBGP.

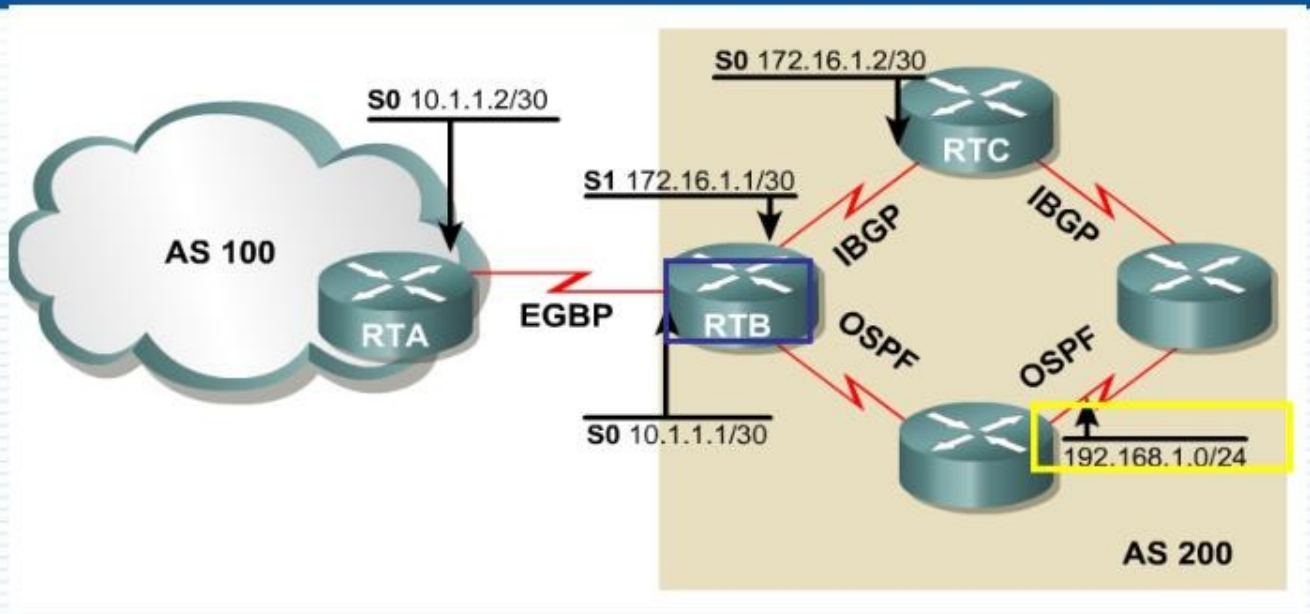
HCL



- Assume the following route appears in RTB's table:

```
0 192.168.1.0/24 [110/74] via 10.2.2.1, 00:31:34,
  Serial2
```

- RTB learned this route via an IGP, in this case, OSPF.
- This AS uses OSPF internally to exchange route information.
- Can RTB advertise this network via BGP?
- Certainly, redistributing OSPF into BGP will do the trick, but the BGP **network** command will do the same thing.

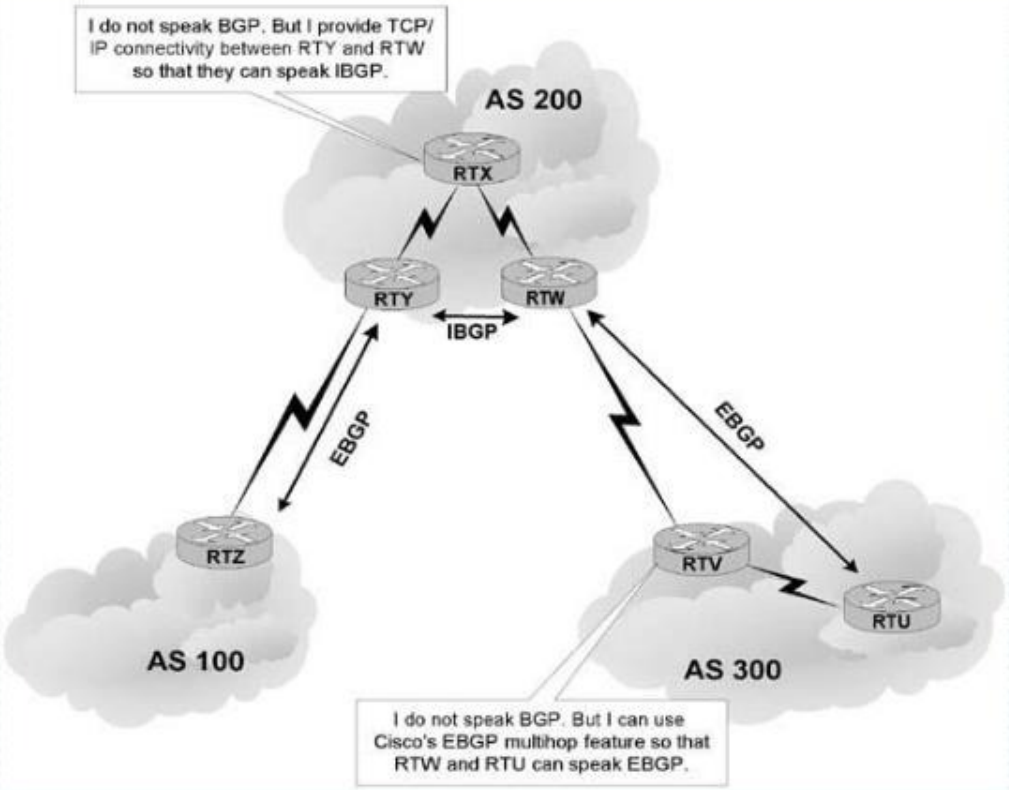



```
RTB(config)#router bgp 200
RTB(config-router)#network 172.16.1.0 mask 255.255.255.254
RTB(config-router)#network 10.1.1.0 mask 255.255.255.254
RTB(config-router)#network 192.168.1.0
```

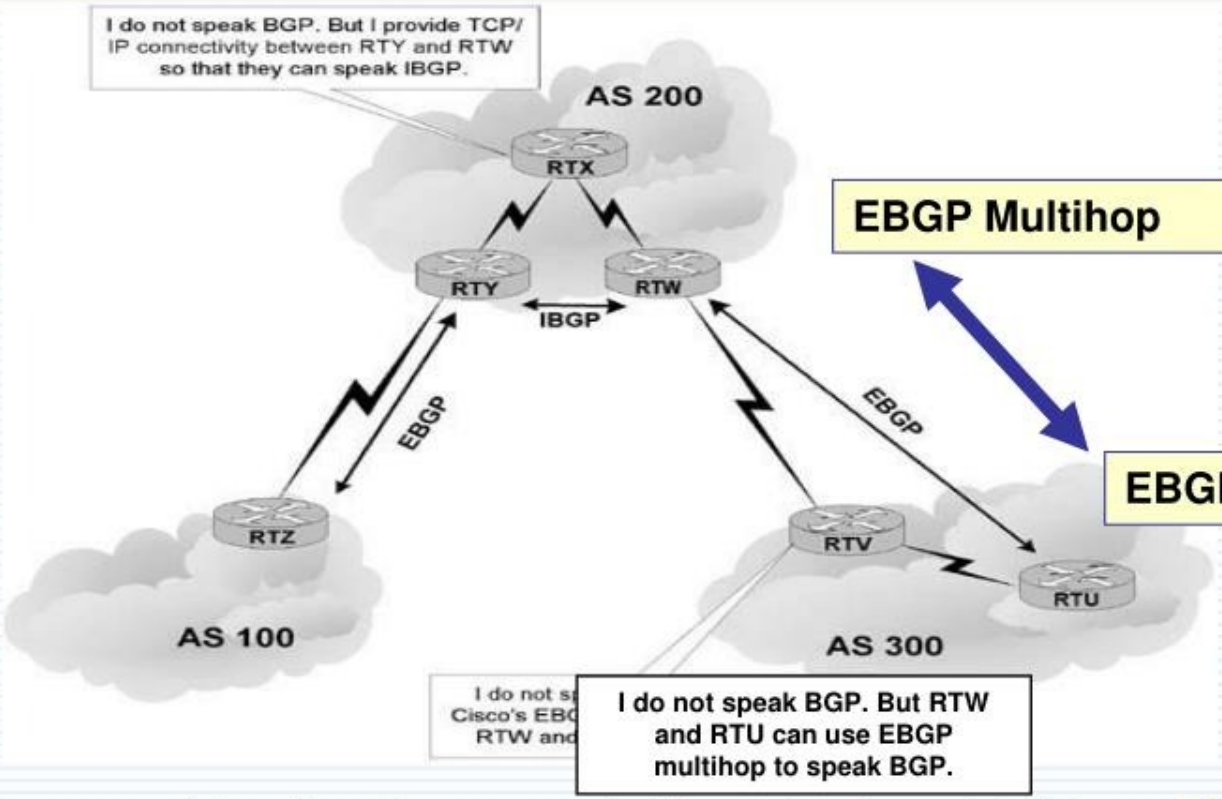
- The first two **network** commands in include the **mask** keyword, so that only a particular subnet is specified.
- The third **network** command results in the OSPF route being advertised by BGP *without* redistribution.
- Remember that the BGP **network** command works differently than the IGP **network** command!

HCL

EBGP vs IBGP



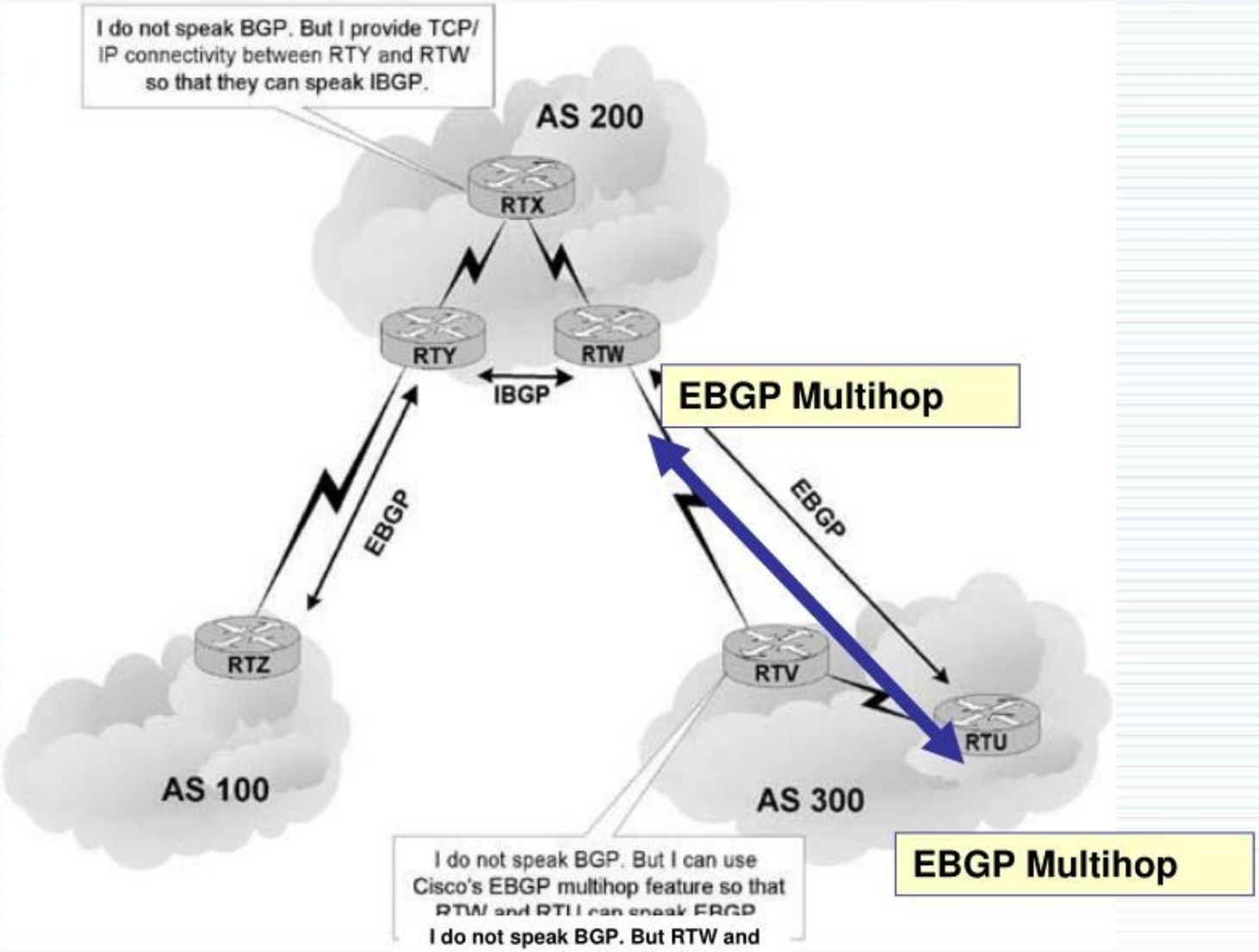
- **EBGP** peers must be **directly connected**, but there are certain exceptions to this requirement.
- In contrast, **IBGP** peers merely **require TCP/IP connectivity** within the same AS.
  - As long as RTY can communicate with RTW using TCP, both routers can establish an IBGP session.
  - If needed, an IGP such as OSPF can provide IBGP peers with routes to each other.



- **EBGP** neighbors must be directly connected in order to establish an **EBGP** session.
- However, **EBGP multihop** is a Cisco IOS option allows RTW and RTU to be logically connected in an **EBGP** session, despite the fact that RTV does not support BGP.
- The **EBGP** multihop option is configured on each peer with the following command:

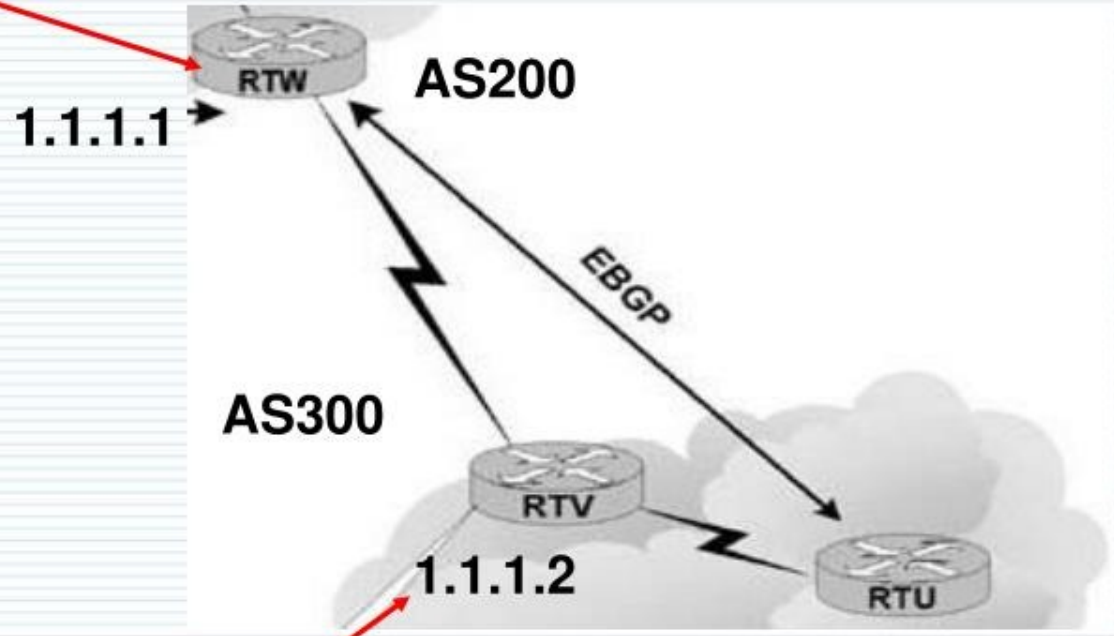
```
Router(config-router)#neighbor IP-address ebgp-multihop
   [hops]
```

EBGP


I do not speak BGP. But I provide TCP/ IP connectivity between RTY and RTW so that they can speak IBGP.
AS 200
RTX
RTY
RTW
IBGP
EBGP Multihop
EBGP
EBGP
RTZ
RTV
RTU
AS 100
AS 300
I do not speak BGP. But I can use Cisco's EBGP multihop feature so that
EBGP Multihop
I do not speak BGP. But RTW and RTU can use EBGP multihop to speak BGP.

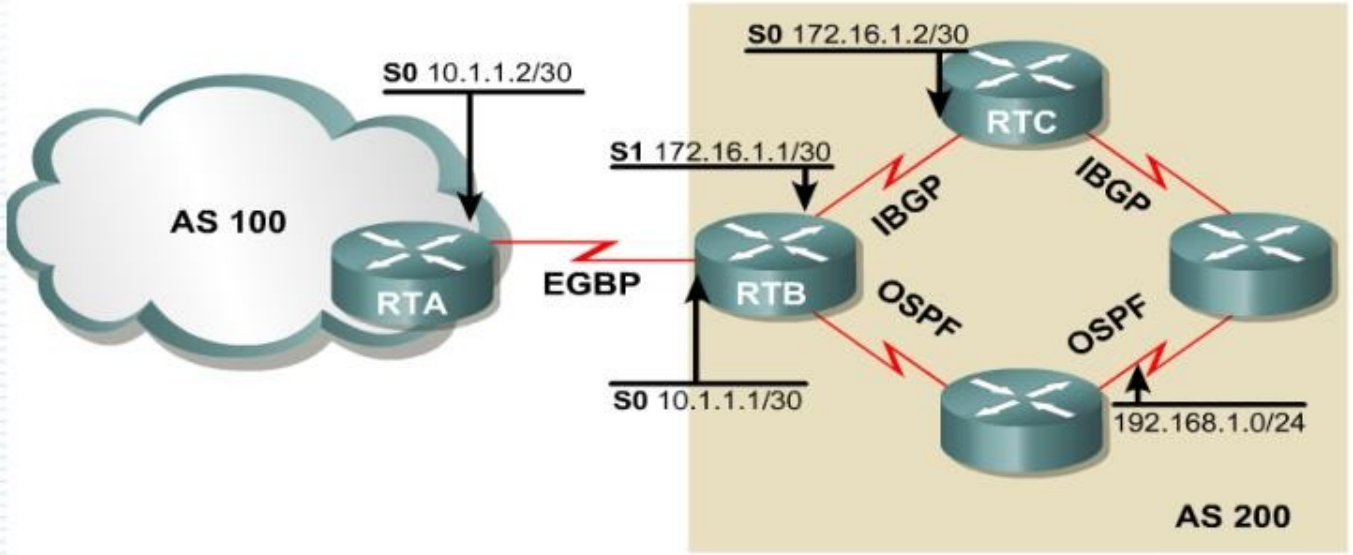
EBGP Multihop

```
RTW(config)#router bgp 200
RTW(config-router)#neighbor 1.1.1.2 remote-as 300
RTW(config-router)#neighbor 1.1.1.2 ebgp-multihop 2
```




```
RTU(config)#router bgp 300
RTU(config-router)#neighbor 1.1.1.1 remote-as 200
RTU(config-router)#neighbor 1.1.1.1 ebgp-multihop 2
```

BGP Configuration



- Finally, whenever you are configuring BGP, you will notice that changes you make to an existing configuration may not appear immediately.
- To force BGP to clear its table and reset BGP sessions, use the **clear ip bgp** command. The easiest way to enter this command is as follows:

```
Router#clear ip bgp *
Router#clear ip bgp 10.0.0.0
```

Use this command with CAUTION, better yet, not at all, in a production network. From the net…

- "clear ip bgp *

Verifying BGP Configuration

- If the router has not installed the BGP routes you expect, you can use the **show ip bgp** command to verify that BGP has learned these routes.

```
RTA#show ip bgp

BGP table version is 3, local router ID is 10.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i -
   internal
Origin codes: i - IGP, e - EGP, ? - incomplete

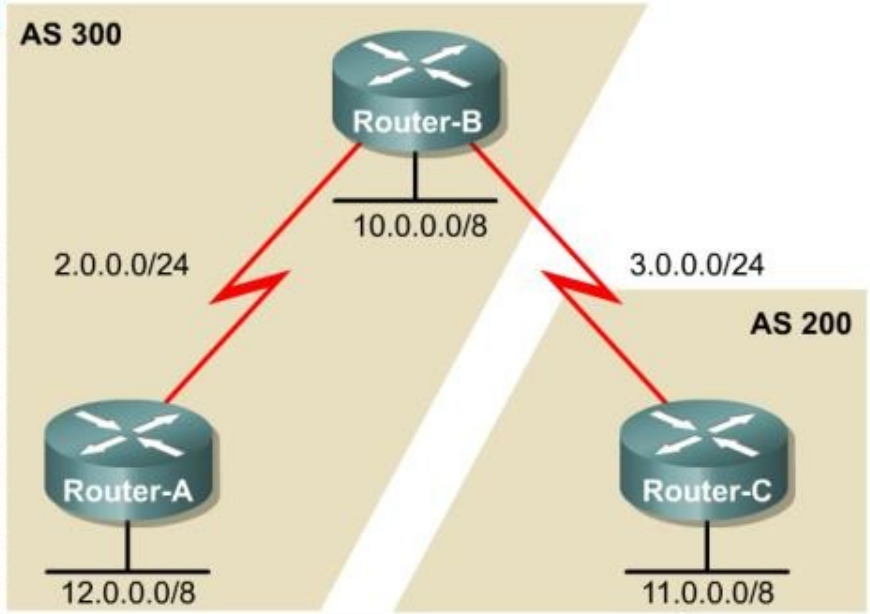
   Network          Next Hop            Metric LocPrf Weight Path
* i1.0.0.0         192.168.1.6              0    100      0 200 400 e
*>i10.1.1.1/32     10.1.1.1                 0    100      0 i
*>i172.16.1.0/24   10.1.1.1                 0    100      0 i
* i192.168.1.32/27 192.168.1.6              0    100      0 200 i
```

Verifying BGP Configuration

- If an expected BGP route does not appear in the BGP table, you can use the **show ip bgp neighbors** command to verify that your router has established a BGP connection with its neighbors.

```
RTA#show ip bgp neighbors
BGP neighbor is 172.24.1.18,  remote AS 200, external link
  BGP version 4, remote router ID 172.16.1.1
  BGP state = Established, up for 00:03:25
  Last read 00:00:25, hold time is 180, keepalive interval is
   60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received
    Address family IPv4 Unicast: advertised and received
  Received 7 messages, 0 notifications, 0 in queue
  Sent 8 messages, 0 notifications, 0 in queue
  Route refresh request: received 0, sent 0
  Minimum time between advertisement runs is 30 seconds
<output omitted>
```

BGP Show Commands



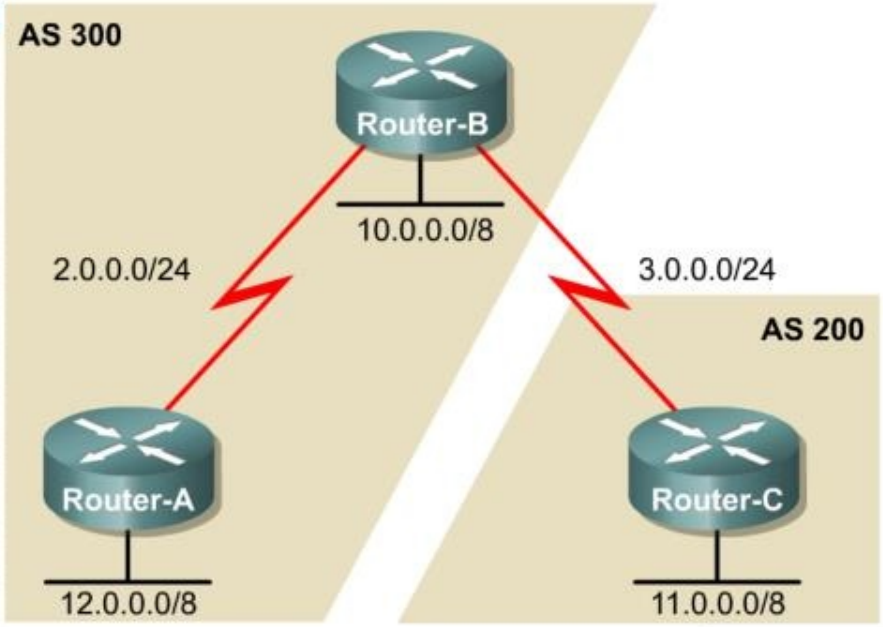

```
Router-A#show ip bgp
BGP table version is 12, local router ID is 12.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - inte
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.0.0.0         2.0.0.1                  0    100      0 i
*>i11.0.0.0         2.0.0.1                  0    100      0 200 i
*> 12.0.0.0         0.0.0.0                  0         32768 i
Router-A#
```

```
Router-A#show ip bgp summary
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/

2.0.0.1         4   300     201     197       12    0    0 00:54:23
Router-A#
```

BGP Show Commands




```
Router-B#show ip bgp
BGP table version is 12, local router ID is 10.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - inte
Origin codes: i - IGP, e - EGP, ? - incomplete

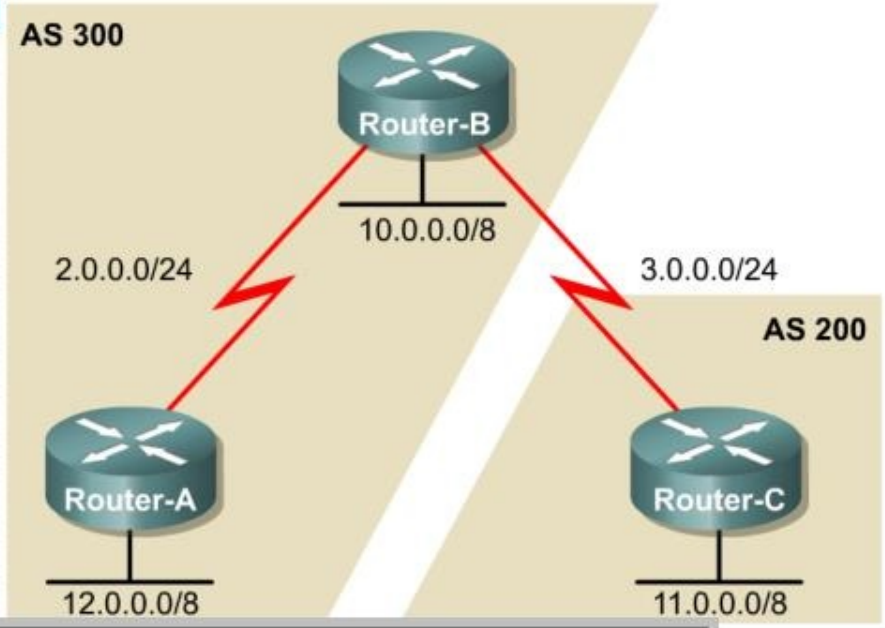
   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.0.0         0.0.0.0                  0         32768 i
*> 11.0.0.0         3.0.0.2                  0             0 200 i
*>i12.0.0.0         2.0.0.2                  0    100      0 i
Router-B#
```

```
Router-B#show ip bgp summary

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/

2.0.0.2         4   300     202     206       12    0    0 00:59:51
3.0.0.2         4   200     172     175       12    0    0 02:17:05
Router-B#
```

BGP Show Commands




```
Router-C#show ip bgp
BGP table version is 6, local router ID is 11.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - inte
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.0.0         3.0.0.1                  0             0 300 i
*> 11.0.0.0         0.0.0.0                  0         32768 i
*> 12.0.0.0         3.0.0.1                                0 300 i
Router-C#
```

```
Router-C#show ip bgp summary
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/

3.0.0.1         4   300     178     176        6    0    0 02:20:15
Router-C#
```

show ip bgp

- RouterC#**show ip bgp**
- **BGP table version** is 8, **local router ID** is 200.200.200.66
- **Status codes**: s suppressed, d damped, h history, * valid, > best, i - internal
- Origin codes: i - IGP, e - EGP, ? - incomplete

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	11.0.0.0	0.0.0.0	0		32768	i
*>	12.0.0.0	200.200.200.65			0	300 200 i
*>	192.10.2.0	200.200.200.65	0		0	300 i

- **BGP table version** - Internal version number of the table. This number is incremented whenever the table changes.
- **local router ID** - IP address of the router.
- **Status codes** - Status of the table entry. The status is displayed at the beginning of each line in the table. It can be one of the following values:

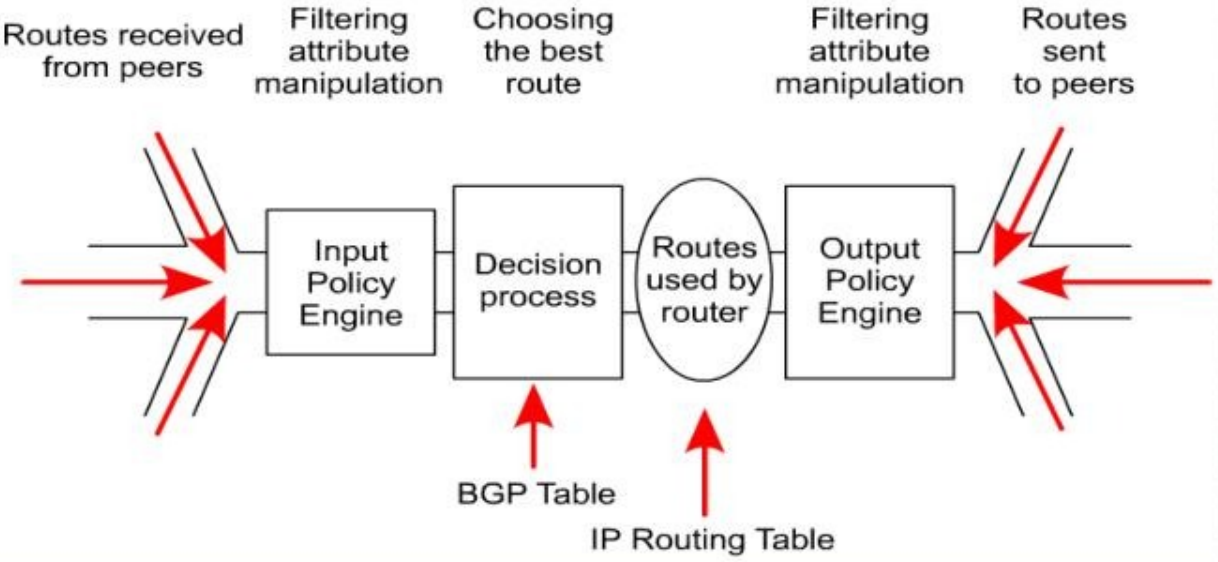
  s —The table entry is suppressed.

  * —The table entry is valid.

  > —The table entry is the best entry to use for that network.

  i —The table entry was learned via an internal BGP (iBGP) session

BGP Routing



- BGP is so flexible because it is a fairly simple protocol.
- Routes are exchanged between BGP peers via UPDATE messages.
- BGP routers receive the UPDATE messages, run some policies or filters over the updates, and then pass on the routes to other BGP peers.
- The Cisco implementation of BGP keeps track of all BGP updates in a BGP table separate from the IP routing table.

BGP Route Selection Algorithm

Summary of the BGP Path Selection Process

- BGP selects only one path as the best path.
- When the path is selected, BGP puts the selected path in its routing table and propagates the path to its neighbors.
- BGP uses the following criteria, in the order presented, to select a path for a destination:

1. If the path specifies a next hop that is inaccessible, drop the update

2. Prefer the path with the **largest weight**.

3. If the weights are the same, prefer the path with the **largest local preference**.

4. If the local preferences are the same, prefer the **path that was originated by BGP** running on this router.

5. If no route was originated, prefer the route that has the **shortest AS_path**.

6. If all paths have the same AS_path length, prefer the path with the **lowest origin** type (where IGP is lower than EGP, and EGP is lower than Incomplete).

7. If the origin codes are the same, prefer the path with the **lowest MED attribute**.

8. If the paths have the same MED, prefer the **external path** over the internal path.

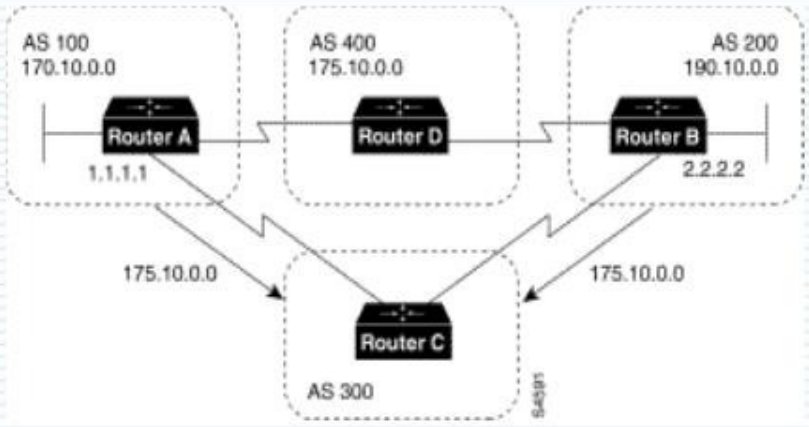
9. If the paths are still the same, prefer the path through the **closest IGP neighbor**.

10. Prefer the path with the lowest IP address, as specified by the BGP **router ID**.

NEXT_HOP

- The **NEXT_HOP** attribute is a well-known mandatory attribute (type code 3).
- In terms of an **IGP**, such as RIP, the “next hop” to reach a route is the IP address of the router that has announced the route.
- The next hop is the IP address of the EBGP neighbor from which the route was learned.

The WEIGHT attribute



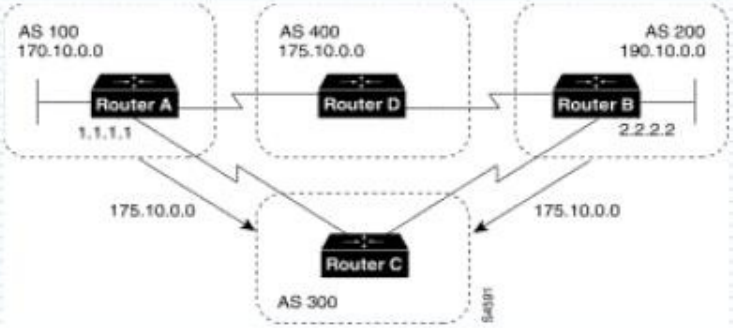
The weight attribute is a **special Cisco attribute** that is used in the path selection process **when there is more than one route to the same destination**.

The weight attribute is **local to the router on which it is assigned**, and it is **not propagated** in routing updates.

By **default**, the weight attribute is **32768** for paths that the router originates and **zero** for other paths.

Routes with a **higher weight are preferred** when there are multiple routes to the same destination.

The WEIGHT attribute



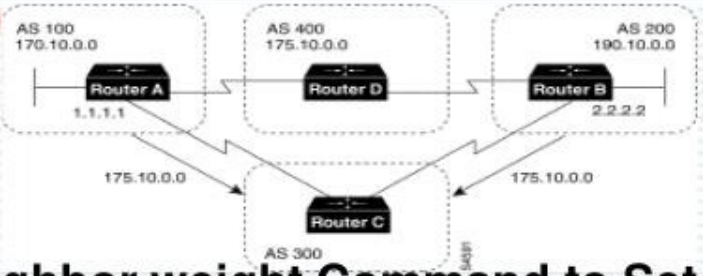
There are three ways to set the weight for updates coming in from Router A:

- Using the **neighbor weight** Command to Set the Weight Attribute
- Using an **Access List** to Set the Weight Attribute
- Using a **Route Map** to Set the Weight Attribute

The Weight Attribute

weight 200

Higher weight preferred



weight 100

Using the neighbor weight Command to Set the Weight Attribute

The following configuration for Router C uses the **neighbor weight** router configuration command:

Router C

```
router bgp 300
   neighbor 1.1.1.1 remote-as 100
   neighbor 1.1.1.1 weight 200
   neighbor 2.2.2.2 remote-as 200
   neighbor 2.2.2.2 weight 100
```

This configuration sets the **weight** of **all route updates** **from AS 100 to 200**, and the **weight** of **all route updates coming from AS 200 to 100**.

Result: The higher **weight** assigned to route updates from AS 100 causes **Router C** to send traffic through **Router A**.

Regular Expressions

- A **regular expression** is a pattern to match against an input string.

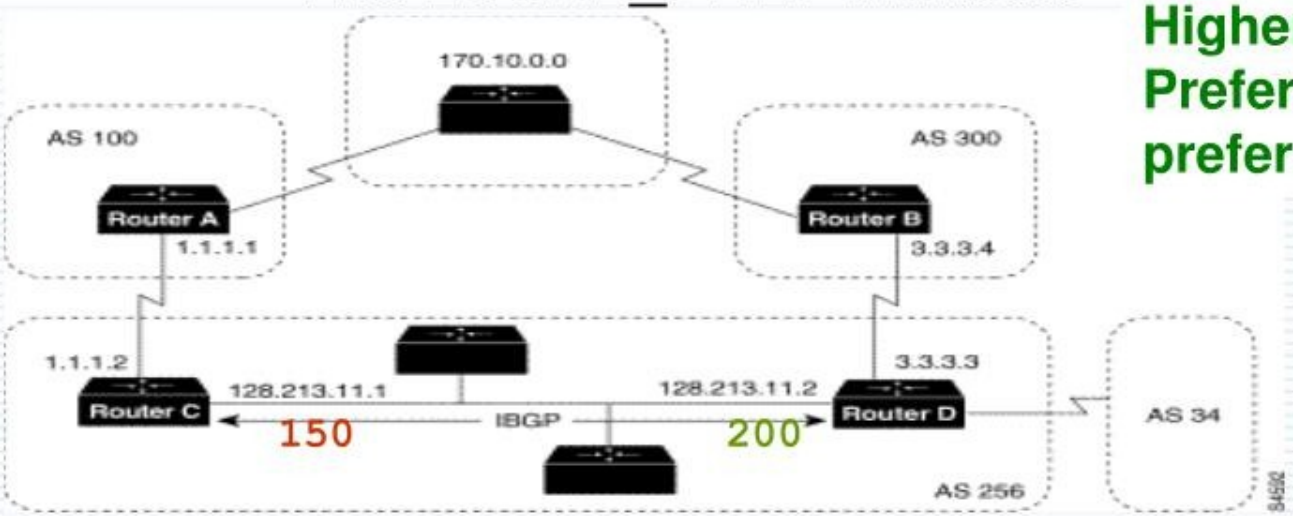
Character	Description
^	Matches the beginning of the input string.
$	Matches the end of the input string.
_	Matches a space, comma, left brace, right brace, the beginning of an input string, or the ending of an input stream
.	Matches any single character
*	Matches 0 or more single- or multiple-character patterns.

The LOCAL_PREF Attribute

- Well-known discretionary attribute (type code 5).

- Degree of preference given to a route to compare it with other routes for the same destination
  - Higher LOCAL_PREF values are preferred
- **Local to the AS**
  - Exchanged between IBGP peers only
  - It is not advertised to EBGP peers
- The path with the **higher preference is preferred** (the **default** value of the **local preference** attribute is **100**).

The LOCAL_PREF Attribute



Higher Local Preference is preferred!

Using the bgp default local-preference Command

The following configurations use the bgp default local-preference router configuration command to set the local preference attribute on Routers C and D:

Router C

```
router bgp 256
  neighbor 1.1.1.1 remote-as 100
  neighbor 128.213.11.2 remote-as 256
  bgp default local-preference 150
```

Router D

```
router bgp 256
  neighbor 3.3.3.4 remote-as 300
  neighbor 128.213.11.1 remote-as 256
  bgp default local-preference 200
```

The MED attribute

- The MULTI_EXIT_DISC (Multi-Exit Discriminator) attribute is an optional non-transitive attribute (type code 4).
- **Informs external neighbors about the preferred path** into an AS that has multiple entry points.
- A lower MULTI_EXIT_DISC (or MED) is preferred over a higher MED.
- The **default** value of the **MED** attribute is **0**.

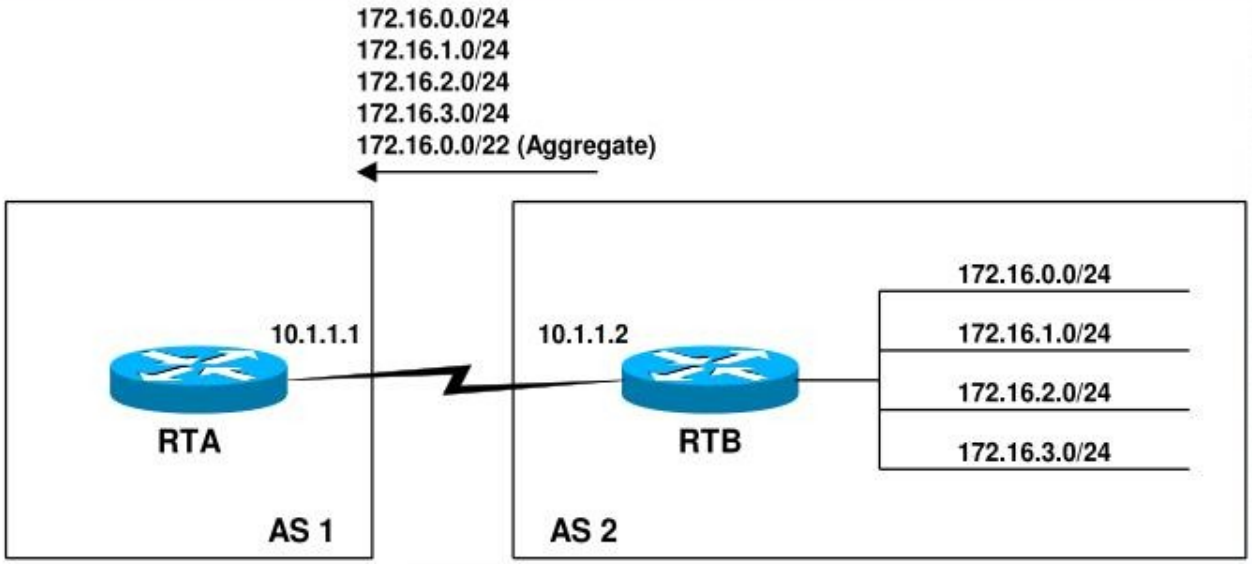
ATOMIC_AGGREGATE

- The **ATOMIC_AGGREGATE** is a well-know discretionary attribute (type code 6).
- The **ATOMIC_AGGREGATE** attribute is set to either "**True**" or "**False**."

You can manually configure BGP to summarize routes by using the **aggregate-address** command, which has the following syntax:

```
Router(config-router)#aggregate-address address mask
   [as-set][summary-only] [suppress-map map-
   name][advertise-map map-name] [attribute-map map-
   name]
```

HCL



RTA

```
router bgp 1
  neighbor 10.1.1.2 remote-as 2
```

RTB

```
router bgp 2
  neighbor 10.1.1.1 remote-as 1
  network 172.16.0.0 mask {/24}
  network 172.16.1.0 mask {/24}
  network 172.16.2.0 mask {/24}
  network 172.16.3.0 mask {/24}
aggregate-address 172.16.0.0
   255.255.252.0 {/22}
```

An aggregate is created only if a more-specific route to the aggregate exists in the BGP table.

BGP Route Filtering

- Configuring a distribute list relies on creating an access list.
- **A standard access list would not work** because it permits more than is desired, since it filters based on the network address only.
- For example, this access list would permit not only the 172.16.0.0/16 summary, but also all the components of that summary as well:

```
access-list 1 permit 172.16.0.0 0.0.255.255
```

BGP Route Filtering - Prefix lists

- **Improved user-friendliness** was one of the factors that motivated Cisco to include the **ip prefix-list** command in IOS 12.0.
- You can use prefix lists as an alternative to access lists with many BGP route-filtering commands.
- There is an **implicit deny at the end of each prefix list**.
- To define a prefix list, use the **ip prefix-list command**, which has the following syntax:

```
Router(config)#ip prefix-list list-name [seq seq-value] deny|permit network/len [ge ge-value] [le le-value]
```

BGP Route Filtering - Prefix lists

Parameter	Description
list-name	Specifies the name of a prefix list.
`seq`	(Optional) Applies the sequence number to the prefix list entry being created or deleted.
seq-value	(Optional) Specifies the sequence number for the prefix list entry.
`deny`	Denies access to matching conditions.
permit	Permits access for matching conditions.
*network/**len***	(Mandatory) The network number and length (in bits) of the network mask.
`ge`	(Optional) Applies *ge-value* to the range specified.
ge-value	(Optional) Specifies the lesser value of a range (the "from" portion of the range description).
`le`	(Optional) Applies *le-value* to the range specified.
le-value	(Optional) Specifies the greater value of a range (the "to" portion of the range description).

BGP Route Filtering - Prefix lists

- Each prefix list entry is assigned a sequence number, either by default or manually by an administrator.
- By numbering the prefix list statements, new entries can be inserted at any point in the list, which is important because routers test for prefix list matches from lowest sequence number to highest.
- By default, the entries of a prefix-list will have sequence values of 5,10, 15, etc.
- To disable this: RTR(config)# no ip prefix-list sequence-number
- Sequence numbers can be created using the command:

```
Router(config)#ip prefix-list list-name [seq seq-
  value] deny|permit network/len [ge ge-value] [le
  le-value]
```

Example

```
RTA(config)#ip prefix-list HCL permit 172.16.0.0/16
RTA(config)#router bgp 100
RTA(config-router)#neighbor 192.168.1.1 remote-as 200
RTA(cnfig-router)#neighbor 192.168.1.1 prefix-list HCL out
```

- The keywords **ge** and **le** can be used to specify the range of the **prefix length** to be matched for prefixes that are more specific than the *network/len* value.
- For example, to accept a mask length of up to 24 bits in **routes** with the prefix 192.0.0.0/8, (ie.192.1.0.0/16, 192.2.10.0/24) and deny more specific routes (192.168.10.128/25), use the commands as shown in.

```
RTA(config)#ip prefix-list HCL permit 192.0.0.0/8 le 24
RTA(config)#ip prefix-list HCL deny 192.0.0.0/8 ge 25
RTA#show ip prefix-list
ip prefix-list HCL: 2 entries
   seq 5 deny 192.0.0.0/8 ge 25
   seq 10 permit 192.168.0.0/8 le 24
```

THANK YOU